When Does Training on Downscaled Images Yield the Same Gradients?

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Abstract

Diffusion transformers deliver strong image generation, but their training cost grows superlinearly with resolution. A recent family of methods therefore trains or samples at reduced resolution on a spectral premise: at high noise, a downscaled latent preserves almost the full surviving signal. The evidence for that premise, however, lives at the level of generated output: whether a downscaled step also delivers the *native training gradient*, under the same model and objective, has not been measured, and existing methods sidestep the question by retraining against stage-specific targets. We measure it directly, with a debiased gap estimator (naive finite-draw estimation manufactures the very effect under test), and find the substitution's cost splits into two parts: a noise-dependent term governed by the downscale *ratio*, and a σ-independent floor governed by the target grid's *absolute token count*, carried by the compute graph itself. No noise level rescues a 2× downscale; but on the mildest route (1024→896) the training gradient is indistinguishable from native at instrument resolution for $\sigma > 0.5$, despite roughly a quarter fewer input tokens: a window that no spectral tolerance predicts, and that our two-term account predicts on held-out routes at 0.07–0.09 RMSE. Gating substitution on that window gives *selective* low-resolution training: −14.6% wall-clock at a fixed step budget, with a sample-level footprint within the training-seed lottery (quality non-inferiority at production scale remains open).

1 Introduction

The cost of a diffusion-transformer forward pass grows superlinearly with resolution: token count grows quadratically with edge length, and attention cost grows again in token count. That makes the title question worth asking precisely. At high noise, can the *gradient* of an adapter be computed on a downscaled latent as a drop-in substitute for the native one, with the same model, the same objective, and the same σ distribution, only the spatial grid changed on some steps?

There is a well-studied reason to hope the answer is yes. As the noise level σ grows, per-frequency signal-to-noise falls, and high spatial frequencies drop below the noise floor first, so that above a spectral crossover a downscaled latent carries *almost* the full surviving signal. But the studies of this resolution gap live almost entirely on the *inference* side: scale-wise distillation (Starodubcev et al., 2025), spectral progressive diffusion (Xiao et al., 2026), and spectrally-guided noise schedules (Esteves & Makadia, 2026) run high-noise *sampling* steps at reduced resolution and judge the premise by the generated output. And where resolution is reduced during training (pyramidal flow matching, Jin et al., 2024; progressive-resolution training more broadly, Karras et al., 2018; Chen et al., 2024; Hoogeboom et al., 2023), the low-resolution stage is given its own, modified objective, and success is again read off the final samples. By the spectral premise, substitution is safe *whenever the noise has masked the frequencies the coarse grid loses*.

None of this work, however, analyzes the gap at *training* time, that is, whether a downscaled step delivers the native gradient under an unchanged objective; that is the route this paper takes. What a training step consumes is the expected adapter gradient, and *demotion* — our term for the training-time substitution of a downscaled, re-encoded latent for the native one — perturbs more than the network's input: the regression target carries the clean image at unit weight at every noise level, and the compute graph itself changes with the grid. Nor does output-level evidence settle the matter: a small quality-score difference does not transfer to a small difference in what a training step *learns*. This paper measures the question directly, at the gradient level, holding model and objective fixed.

Measuring it honestly is, unexpectedly, a metrology problem. The natural gap estimator (a redraw floor minus a single demoted-arm cosine) is biased by finite-draw variance, and the bias grows as the demoted grid shrinks: it *manufactures* exactly the token-count-ordered floor signature under test. Every claim in this paper is therefore stated in debiased units, at the instrument's measured resolution.

The contributions:

1. **Gap metrology and a two-term account of what demotion costs** (§3). We define the demotion gap at the gradient level and build a debiased instrument for it: per-arm self-floors with a split-half attenuation correction, a draw-count extrapolation $\mathrm{gap}(D) = \mathrm{gap}_\infty + c/D$, and a certification resolution $\varepsilon^*(N, D)$ under which "safe" is a one-sided non-inferiority statement, not an absolute claim. From the perturbation expansion and cosine geometry, the gap reduces under four named assumptions to a *two-term account*: a route amplitude on a universal mismatch-amplitude curve read through the U-shaped gradient norm, plus a σ-independent graph floor. The account connects directly to the inference-side literature: the spectral argument, in its strongest tolerance-parameterized form (SPD, Xiao et al., 2026), yields its own account of the data term's input-mediated part, one that is ratio-only and floorless by construction.
2. **A measured safety map and a predictive tool** (§4, §5). Measured with that instrument, the mildest route (1024→896) is indistinguishable from the re-encode control for $\sigma > 0.5$ at instrument resolution, while no noise level rescues a 2× downscale, and no member of the spectral tolerance family reproduces the measured map. The two-term account *predicts* held-out routes at ∼0.07–0.09 RMSE, with the floor law correctly predicting both held-out floors: a tool for anticipating a route's gap before measuring it. Gating substitution on the measured-safe windows gives an opt-in σ-conditional trainer realizing −15.1% forward FLOPs and −14.6% wall-clock at a fixed step budget, with a sample-level footprint within the training-seed lottery (the seed-noise yardstick, Table 4); generation-quality non-inferiority at production scale remains the stated open gate.

We first situate the question in the prior literature (§2); §3 defines the estimand and its certification resolution, derives the angular account of the demotion gap and the gradient gap the spectral argument implies for its data term, and constructs the debiased estimator; §4 builds the instrument and discharges both accounts' predictions against the measurements; §5 gives the resulting safety map and the σ-conditional trainer.

2 Related work

Flow matching and DiT. Flow matching (Lipman et al., 2023) trains a continuous-time generative model by regressing a velocity field along prescribed probability paths between noise and data; rectified flow (Liu et al., 2023) takes the paths to be straight lines, giving the linear noising $z_\sigma = (1 - \sigma)x + \sigma\epsilon$ and the constant regression target $v = \epsilon - x$ used throughout this paper, and this formulation is the training objective of current large text-to-image systems (Esser et al., 2024). The Diffusion Transformer (Peebles & Xie, 2023) replaces the U-Net denoiser with a plain transformer over patchified latent tokens, so a spatial grid enters the network only as a token sequence: token count grows quadratically with edge length and attention cost quadratically again in token count, the cost structure that makes resolution the expensive axis. Spatial position is carried by rotary position embeddings (Su et al., 2024), whose per-band phases accumulate with grid coordinates; this grid-dependent coordinate system is the one whose contribution to the demotion floor §4.5 isolates by intervention. Fine-tuning is LoRA (Hu et al., 2022): low-rank adapter updates on a frozen backbone, the gradient regime all our probes measure.

Scale-wise and progressive-resolution diffusion. Progressive growing (Karras et al., 2018) and resolution curricula (Chen et al., 2024; Hoogeboom et al., 2023) train at increasing resolution as a schedule, each low-resolution stage carrying its own stage-specific objective. A more recent family ties resolution to the noise level itself, on a spectral premise. Under flow-matching noising, each spatial-frequency band of the signal retains an effective signal-to-noise ratio that falls as σ grows, and the finest bands sink below the noise floor first; the noise level above which a band is masked is that band's *crossover* σ. Above the crossover of the bands a coarse grid cannot represent, a downscaled latent should therefore carry almost the full surviving signal. This resolution gap has so far been studied on the *inference* side. Scale-wise distillation (SwD) (Starodubcev et al., 2025) states the claim directly (above the crossover, the noised downscaled latent is close in distribution to the downscaled noised latent) and samples early steps at reduced resolution. Spectral Progressive Diffusion (SPD) (Xiao et al., 2026) gives the argument its sharpest available form: a tolerance-parameterized per-frequency activation time under a diagonal Gaussian spectral model, a criterion on the *Bayes-optimal predictor's* per-band output rather than merely on input SNR, from which an inference-time resolution schedule follows. Spectrally-guided noise schedules (Esteves & Makadia, 2026) apply the same per-frequency reasoning to schedule design, deriving bounds on the useful minimum and maximum noise levels from the image spectrum; here the spectral account operates as the field's working model of what noise masks. Where the premise does enter training, as in pyramidal flow matching (Jin et al., 2024), the objective is restaged per pyramid level rather than left native, and the premise is again validated by the generated output. None of these analyzes what a low-resolution step does to the *training gradient* of an unchanged objective; that is the question we measure, holding model and objective fixed. We find that input-level closeness does not transfer at any tolerance.

Positional extension. Position interpolation (Chen et al., 2023), banded/frequency-selective variants (YaRN, Peng et al., 2024), and diffusion-specific dynamic position extrapolation (Issachar et al., 2026; Zhao et al., 2026) rescale rotary coordinates to evaluate a model at unseen grid sizes. We use exact positional interpolation as a *causal intervention* (matching the downscaled grid's relative phase geometry to native) to decompose the resolution-sensitivity floor, and adopt the σ-gated band schedule of Zhao et al. (2026) as a training-time refinement.

3 Theory: the demotion gap and a two-term account

3.1 Estimand

Objective and noising. We first define the training objective whose gradients are under test. A flow-matching model is trained to predict the constant-velocity transport between a clean latent and pure noise. With clean latent x, noise $\epsilon \sim \mathcal{N}(0, I)$, and a noise level $\sigma \in (0, 1]$, the noised input is the linear interpolation $z_\sigma = (1-\sigma)\,x + \sigma\,\epsilon$ (at $\sigma \to 0$ the input is the clean latent, at $\sigma = 1$ it is pure noise), and the regression target is the velocity $v = \epsilon - x$ that carries z_σ along the interpolation. With caption condition c, the loss is $\|\hat{v}_\theta(z_\sigma, \sigma, c) - v\|^2$, and gradients are taken with respect to the LoRA adapter parameters only; the backbone is frozen.

Demotion. A *demotion* $e_0 {\to} e$ replaces the native latent with a coarser-grid one, produced exactly as a scale-wise trainer would produce it; e_0 denotes the route's source (native) edge and e its target edge. With y the image at its native (tier-e_0) resolution, $\mathcal{E}$ the VAE encoder, and $\mathcal{R}_e$ the pixel-space downscale to the tier-e bucket (native aspect ratio preserved),

$$x_{\text{src}} \;=\; \mathcal{E}(y) \quad\longrightarrow\quad x_{\text{dem}} \;=\; \mathcal{E}\big(\mathcal{R}_e(y)\big),$$

a latent with proportionally fewer spatial tokens; we do not downsample in latent space. Everything else about the training step is held fixed: the caption c, the σ under test, and the noise draw structure (the same stratified draw scheme, with ϵ drawn at the tier-e shape). The demoted step then trains on $z_\sigma = (1-\sigma)\,x_{\text{dem}} + \sigma\,\epsilon$ with target $v = \epsilon - x_{\text{dem}}$, so the substitution changes both what the network sees and what it is asked to predict, but only through the spatial grid; nothing else in the step references the resolution.

The estimand. Three *arms* recur throughout: src, the source arm, which trains on the native-resolution latent x_{src}; dem, the demoted arm, which trains on x_{dem}; and reenc, a control arm that decodes and re-encodes the native latent at native resolution: it pays the VAE round trip that demotion necessarily pays, with no grid change. For an arm $a \in \{\text{src}, \text{reenc}, \text{dem}\}$, let $\bar{g}_a(\sigma) = \mathbb{E}_\epsilon\,[\nabla_\theta \mathcal{L}]$ be the population per-bin mean adapter gradient of a query (image, caption). The *population demotion distance* of the route $e_0 {\to} e$ is

$$d_{e_0\to e}(\sigma) \;=\; 1 - \cos\big(\bar{g}_{\text{src}}(\sigma),\; \bar{g}_{\text{dem}}(\sigma)\big) \;=\; \tfrac{1}{2}\,\big\|\hat{g}_{\text{src}} - \hat{g}_{\text{dem}}\big\|^2, \tag{1}$$

with $\hat{g}$ the unit-normalized gradients; $d_{e_0 \to e} \geq 0$ always. One notational convention holds from here on: arm labels (src, reenc, dem) mark *whose* gradient, latent, or Jacobian an object is; the subscript on every route-level scalar is the *route*, written in full, $d_{e_0 \to e}$, where several sources appear, and abbreviated to the target edge alone — an edge value, never an arm label, so it instantiates numerically (d_e, Floor_{1120}, Resid_{768}) — wherever the source edge is fixed by context, as it is below. The control's distance d_{reenc} — Eq. 1 with $\bar{g}_{\mathrm{reenc}}$ in place of $\bar{g}_{\mathrm{dem}}$ — carries its arm label instead: the control changes no grid and has no target edge. The *reported* quantity of every bin-resolved map and every fit in this paper is the *re-encoding excess* $\mathrm{gap}_e(\sigma) := d_e(\sigma) - d_{\mathrm{reenc}}(\sigma)$, the demotion distance beyond what the control already pays; gap_e can be negative, d_e cannot. Endpoint and floor tables report debiased estimates of d_e with the control as its own row; each table states which object it reports.

In a batched training scenario the same definition admits two aggregations over a probe set, and the aggregation operator is part of the estimand: per-example (cosine per image, then mean) or batch-aggregate (gradients summed over the batch before the cosine; the object SGD actually follows). The two are related by a decomposition, not an ordering. Writing each image's demotion-induced gradient disagreement as a coherent mean b plus a zero-mean idiosyncratic deviation with covariance Σ_η, the batch-aggregate distance at batch size B is, to second order, an intercept plus a $1/B$ term,

$$\mathbb{E}[d_B] \;\approx\; \frac{\|P^\perp b\|^2}{2\|\mu\|^2} \;+\; \frac{\mathrm{tr}\big(P^\perp \Sigma_\eta P^\perp\big)}{2B\|\mu\|^2},$$

with μ the mean source gradient and $P^\perp$ the projector off $\hat{\mu}$: the coherent share never averages out. The per-example map is therefore *not* a lower bound for larger batches — monotone improvement with B would require an iid zero-mean disagreement model we have not verified. These are different estimands, and no coefficient fitted on one is guaranteed to transfer to the other; the application section publishes one map for each (§5.1).

Finally, any estimate of $d_e(\sigma)$ compares finite-draw estimates of population directions, so every claim carries the instrument's resolution. For a route whose true excess is zero, the paired estimate $\overline{\mathrm{gap}}$ fluctuates with standard error $\mathrm{SE}(N, D, \sigma\text{-bin})$; its one-sided 95% upper bound defines the *certification resolution* ε^* (the smallest margin certifiable even for a truly safe route) and "safe at margin ε" as the non-inferiority statement

$$\mathrm{UB}_{95}(\overline{\mathrm{gap}}) \;<\; \varepsilon, \qquad \varepsilon \;\geq\; \varepsilon^*(N, D, \mathrm{bin}) \;=\; 1.645\,\mathrm{SE}(N, D, \mathrm{bin}), \tag{2}$$

not a hard bound. Throughout, "no certifiable window" means that a route's upper bounds do not cross below ε at the current (N, D), while "certifiably unsafe" means that its lower bound stays above it.

3.2 Data and graph perturbations

We now derive the perturbation demotion applies to the gradient. Differentiating the loss of §3.1 in the adapter parameters gives, per draw and up to a constant,

$$g \;=\; \nabla_\theta \tfrac{1}{2}\big\|\hat{v}_\theta - v\big\|^2 \;=\; J^\top r,$$

with $r = \hat{v}_\theta - v$ the prediction residual and $J = \partial \hat{v}_\theta / \partial \theta$ the Jacobian through which the compute graph turns a residual into an adapter gradient. Demotion changes exactly the two factors of this product, and nothing else about the map from (image, noise, caption) to adapter gradient: the residual, because it changes the data the objective consumes, and the Jacobian, because it changes the compute graph the gradient flows through. Averaging over draws and expanding the product of the perturbed factors (an elementary but bookkeeping-heavy step, written out in Appendix A together with what exactly the remainder collects), the perturbation $\delta g = \bar{g}_{\mathrm{dem}} - \bar{g}_{\mathrm{src}}$ decomposes as

$$\delta g \;=\; \underbrace{J^\top \Delta \bar{r}(\sigma)}_{\text{data branch}} \;+\; \underbrace{\Delta J^\top \bar{r}}_{\text{graph branch}} \;+\; \underbrace{\Delta J^\top \Delta \bar{r} + \cdots}_{\text{remainder } R_e}, \tag{3}$$

where:

1. **the data branch** $J^\top \Delta \bar{r}$: demotion changes the *data* the objective consumes; spatial-frequency bands above the coarse grid's Nyquist limit are absent from the input *and* from the target $v = \epsilon - x$, and the re-encode perturbs what survives, yielding a perturbation $\Delta \bar{r}(\sigma)$

of the mean prediction residual. Its input-mediated part shrinks as the model's reliance on input detail shrinks with σ; its target-mediated part does not: perturbing the image by δx perturbs the residual by $D_z f\,[(1-\sigma)\,\delta x] + \delta x$, and the second term carries unit weight at every σ. At σ=1 the input is pure noise on every grid, so the input-mediated part vanishes in distribution; the endpoint is nevertheless *not* data-free by construction, and whether the surviving target-mediated share is resolvable is an empirical question, discharged in §4.3 (it is real and large, but demotion perturbs it almost entirely *parallel* to the native gradient, so the angular estimand does not see it).

2. **the graph branch** $\Delta J^\top \bar{r}$: token count, rotary phase density, and sequence-length-dependent normalization, collected as an operator perturbation ΔJ that is σ*-independent*, because the grid change does not reference σ.

One qualification is worth stating precisely: J_{src} and J_{dem} act on different grids, so the two branches are not literal matrix products across arms; they are defined *operationally*, as the two independent perturbation directions in the shared adapter-parameter space where both gradients live, isolated by the designated probes of §4.3.

This decomposition already shows why input-level spectral sufficiency cannot bound the gradient. Input noise masking alone provides no bound on expected-gradient disagreement, because systematic target and graph terms survive noise averaging: the quantity training consumes is $\mathbb{E}_\epsilon[\nabla_\theta \mathcal{L}]$, the target carries the clean image at coefficient one at every σ, and ΔJ does not reference the spectrum at all. Input-level sufficiency licenses inference-time substitution; it does not extend to training-gradient equivalence.

3.3 Angular geometry: the exact form and its expansions

It remains to read the perturbation of Eq. 3 in the units of the estimand, Eq. 1: a pure exercise in cosine geometry, whose forms below are what every measurement in §4 is fit against (step-by-step derivation in Appendix B). Writing $u^\perp = u - (\hat{g}_{\text{src}}^\top u)\,\hat{g}_{\text{src}}$ for the component of a vector orthogonal to $\bar{g}_{\text{src}}(\sigma)$, split the perturbation into a rescaling and a rotation: $\kappa_\parallel = \hat{g}_{\text{src}}^\top \delta g \,/\, \|\bar{g}_{\text{src}}\|$ and $\kappa_\perp = \|\delta g^\perp\| \,/\, \|\bar{g}_{\text{src}}\|$. Substituting $\bar{g}_{\text{dem}} = \bar{g}_{\text{src}} + \delta g$ into Eq. 1 then gives, *exactly*,

$$d_e \;=\; 1 - \frac{1}{\sqrt{1+\kappa_{\text{eff}}^2}}\,, \qquad \kappa_{\text{eff}} \;=\; \frac{\kappa_\perp}{1+\kappa_\parallel}\;: \tag{4}$$

the gap is a saturating function of a single rotation-to-scale ratio: only the orthogonal component rotates the gradient direction, and a parallel component merely renormalizes it. Taylor-expanding Eq. 4 at small κ ($d_e = \frac{1}{2}\|\delta g^\perp\|^2/\|\bar{g}_{\text{src}}\|^2 + \cdots$) and substituting the branch decomposition of Eq. 3 into the square gives the *four-term angular expansion*, which, like Eq. 4, is derived unconditionally:

$$d_e(\sigma) \;\approx\; \underbrace{\frac{\big\|(J^\top \Delta\bar{r})^\perp\big\|^2}{2\,\|\bar{g}_{\text{src}}\|^2}}_{\text{data share } S_e} + \underbrace{\frac{\big\|(\Delta J^\top \bar{r})^\perp\big\|^2}{2\,\|\bar{g}_{\text{src}}\|^2}}_{\text{graph share } F_e} + \underbrace{\frac{\big\langle (J^\top \Delta\bar{r})^\perp,\,(\Delta J^\top \bar{r})^\perp\big\rangle}{\|\bar{g}_{\text{src}}\|^2}}_{\text{projected interaction } I_e} + R'_e, \tag{5}$$

where R'_e now has an exact characterization (Appendix B): it collects the remainder branch R_e, the beyond-quadratic terms of Eq. 4, and the parallel-component correction. S_e and F_e are non-negative; I_e can carry either sign, a fact that matters in §4.6. The expansion's validity domain deserves stating where it is introduced: at the harshest measured operating point ($d = 0.30$ at 512, σ=1) the implied perturbation is $\kappa_{\text{eff}} = \sqrt{(1-d)^{-2}-1} \approx 1.02$ — not small — so quadratic-order reads on that route are indicative, and the exact form (Eq. 4) carries the quantitative claims.

From the expansion to the excess. The reported object is the excess over the control (§3.1): let $\text{gap}_e(\sigma) = d_e(\sigma) - d_{\text{reenc}}(\sigma)$. The control's own expansion is degenerate: the reenc arm changes no grid, so $\Delta J_{\text{reenc}} = 0$, and both control terms built from it vanish identically — its graph share F_{reenc} and its projected interaction I_{reenc} each carry a factor $(\Delta J_{\text{reenc}}^\top \bar{r})^\perp = 0$ — leaving only the re-encode perturbation in its data branch. Writing Eq. 5 for each arm and subtracting term by term,

$$\begin{aligned}
\mathrm{gap}_e(\sigma) &= d_e(\sigma) - d_{\mathrm{reenc}}(\sigma) \\
&\approx \big(S_e - S_{\mathrm{reenc}}\big) + \big(F_e - F_{\mathrm{reenc}}\big) + \big(I_e - I_{\mathrm{reenc}}\big) + \big(R'_e - R'_{\mathrm{reenc}}\big) && \text{Eq. 5, each arm} \\
&= \big(S_e - S_{\mathrm{reenc}}\big) + F_e + I_e + \big(R'_e - R'_{\mathrm{reenc}}\big), && F_{\mathrm{reenc}} = I_{\mathrm{reenc}} = 0
\end{aligned}$$

so the control collapses to $d_{\mathrm{reenc}} \approx S_{\mathrm{reenc}} + R'_{\mathrm{reenc}}$, with S_{reenc} the re-encode share (measured ≈ 0 where probed: the control row of Table 3). The subtraction cancels only the re-encode component of the data share (up to a cross term suppressed by $\sqrt{S_{\mathrm{reenc}}}$, bounded by the control's near-zero reading), so m is henceforth read net of re-encode; the graph share and the interaction pass through untouched — the control has nothing to cancel them with. Disposing of the surviving terms is what the assumptions below are for.

The reduced two-term account. The subtraction form above is exact bookkeeping, but not yet an instrument: none of its four terms is separately measurable per bin. Four named assumptions — each tested by a designated probe in §4 — trade it for a form built from measurable curves and per-route constants: (i) **small remainder**: R'_e (and its control counterpart) negligible over the claimed domain; (ii) **negligible projected interaction**: $|I_e| \ll S_e + F_e$ over the claimed domain; (iii) **graph-relative stationarity**: $\|(\Delta J^\top \bar{r})^\perp\| / \|\bar{g}_{\mathrm{src}}\|$ is σ-constant (numerator and denominator share the residual's scale), so the graph share F_e is flat in σ; write Floor_e for its constant value; (iv) **data-branch factorization**: $\|(J^\top \Delta\bar{r})^\perp\| \approx a_e\, m(\sigma)$, a σ-independent route amplitude times the route-independent mismatch shape $m(\sigma) = \|\Delta\bar{r}(\sigma)\|$ of the model's mean prediction residual across grids (directly measurable, §4.4; read net of re-encode per the subtraction above). Note m is a *norm*: the factorization assumes a route-independent amplitude curve, not a shared mismatch *direction* — and §4.4 finds the directions are in fact image- and route-specific. Applied in that order,

$$\begin{aligned}
\mathrm{gap}_e(\sigma) &\approx \big(S_e - S_{\mathrm{reenc}}\big) + F_e && \text{by (i), (ii)} \\
&\approx \underbrace{A_e \cdot \big(m(\sigma)/\|\bar{g}_{\mathrm{src}}\|\big)^2}_{\text{data term}} + \underbrace{\mathrm{Floor}_e}_{\text{graph term}}, \qquad A_e = \tfrac{1}{2}\, a_e^2, && \text{by (iii), (iv)} \qquad (6)
\end{aligned}$$

the *two-term account*, stated on the reported form of the estimand, the re-encoding excess of §3.1, with $\|\bar{g}_{\mathrm{src}}(\sigma)\|$ the total gradient norm. The quadratic power on $m/\|\bar{g}_{\mathrm{src}}\|$ is fixed by the cosine geometry *locally*; far from the small-mismatch regime the licensed read is the parent form itself: keeping Eq. 4 un-expanded, with the factorization's loading written as $\kappa_e(\sigma) = c_e\, m(\sigma)/\|\bar{g}_{\mathrm{src}}(\sigma)\|$ (so $A_e = c_e^2/2$ in the small-κ limit), yields a data term that *saturates* instead of overshooting when κ is comparable to one. The loading itself (that the orthogonal gradient mismatch is proportional to the residual mismatch, $\|\delta g^\perp\| = c_e\, m(\sigma)$) is an *empirical* ingredient, not a derivation; so wherever a fitted functional form is reported, the un-expanded Eq. 4 is labeled the *best empirical predictor* and Eq. 6 the *derived small-perturbation form*; §4.6 scores both, held out.

The graph term has identifiable parts. Changing the grid changes (i) the rotary coordinate system, in which every band of the position embedding accumulates phase at a different density across the image, and (ii) non-positional graph statistics: attention softmax over a different token count, sequence-length-dependent normalization, and the sheer capacity of the coarse graph to approximate the fine graph's computation (the conceptual frame is attention as a discretization of a function-space operator, Calvello et al., 2024, whose statistics need not be discretization-invariant; the frame supplies no quantitative law for the dependence). Splitting the graph share $\mathrm{Floor}_e = \mathrm{RoPE}_e + \mathrm{Resid}_e$ yields one sharp prediction: the positional share is a *coordinate-system artifact*, so an exact positional-interpolation intervention (evaluating the demoted grid at fractional rotary coordinates that reproduce the native relative phase geometry) should remove RoPE_e wherever the data branch is inactive ($\sigma \to 1$); and a band-counting sketch says the residue should grow with absolute grid change (the number of rotary bands whose per-extent rotation count crosses decorrelation grows with the extent), attaching an absolute-size governor to Resid_e. We flag the sketch's epistemic status plainly: like YaRN's own band thresholds (Peng et al., 2024), its constants are calibrated, not derived; we claim no closed form for Floor_e.

The ledger. Explicitly: the exact form (Eq. 4) and the four-term expansion (Eq. 5) are derived unconditionally from Eqs. 1 and 3 (Appendix B); the two-term reduction (Eq. 6) is derived *conditional* *on the four named assumptions*, each of which is tested by a designated probe (§4); the coefficients A_e (or c_e), $m(\sigma)$, $\|\bar{g}_{\text{src}}(\sigma)\|$, Floor_e are measured, not derived; and any fitted power or loading is labeled empirical. The account was pre-registered with an outcome matrix (including a kill switch: all endpoint gaps ≈ 0 would have falsified the graph term) before the discriminating runs.

3.4 The spectral account of the data term, from prior work

The spectral argument reviewed in §2 supplies a second reading of the same gap, and it is worth deriving in the units the measurements will be stated in. Its strongest available form is SPD's tolerance-parameterized crossover (Xiao et al., 2026): under the premise that each clean-latent frequency ω is Gaussian, $x_0^{(\omega)} \sim \mathcal{N}(0, P_\omega)$ (with P the clean-latent power spectrum), their Proposition 1 states that for all $\sigma \geq t_\omega$ the *Bayes-optimal* per-band velocity prediction is within tolerance δ of pure noise, $\mathbb{E}[|v^{*(\omega)} - \epsilon^{(\omega)}|^2] \leq \delta$, and their Proposition 2 places the safe-substitution boundary for a coarse grid at the t_ω of its Nyquist band ω_e. Carrying this to the training gradient — under the premise that demotion costs only what the destroyed bands still contribute to the prediction, so that in the terms of Eq. 5 the gap is the data share alone, gated at the Nyquist band's activation time, with no target-mediated share, no graph share, and no interaction — the gradient gap the argument implies is

$$\text{gap}_e^{\text{spec}}(\sigma) \;=\; S_e(\sigma)\,\mathbf{1}\big[\sigma < t_{\omega_e}\big], \qquad t_\omega \;=\; \frac{1}{1 + \sqrt{\delta \,/\, \big(P_\omega\,(1 + P_\omega - \delta)\big)}}, \qquad F_e = I_e = 0. \quad (7)$$

This is a prediction-level criterion, strictly stronger than raw input SNR; the often-quoted equal-power crossover $\sigma_{\text{eq}}(f) = \sqrt{P(f)}/(1 + \sqrt{P(f)})$ is its $\delta = (1 + P_\omega)/2$ slice.

Three structural consequences are fixed by the form of Eq. 7 before any gradient is measured. All route boundaries move together under the single tolerance δ: the family is totally ordered by one parameter. In normalized frequency the demoted grid's Nyquist sits at $f_{\text{cut}} = \frac{1}{2}\,e/e_0$, a function of the downscale *ratio* alone, so the account cannot distinguish routes of equal ratio at different absolute sizes and is structurally blind to any absolute-size effect. And for every route and every δ the predicted gap vanishes above a finite boundary: every route becomes safe at some σ, and no σ-independent floor is expressible. Under the same diagonal-Gaussian premise the account also extends to the residual level: the posterior mean is a per-band Wiener shrinkage (under a Gaussian prior and Gaussian noise the minimum-mean-square-error estimator is linear with per-band gain $P_\omega/(P_\omega + n_\omega)$, the classical Wiener filter, Wiener, 1949), so with no cross-frequency coupling the cross-grid mean-residual mismatch is confined to the destroyed band and ordered across routes by destroyed-band energy.

The family is instantiated on our measured latent spectrum in §4.2 (Table 2); its curve-level read in gap units necessarily borrows our bridge: the derivation emits residual units only, so any gap-unit prediction is *the spectral account transported through* $\|\bar{g}_{\text{src}}\|$ *and a calibrated route gain*, not a parameter-free prediction (§4.6). Two properties of the instantiation are fixed before any gradient is measured: the latents are high-frequency-quiet, so the family is *compressed* (the spread between the mildest and harshest route remains below ~0.13 in σ at every tolerance), and the two routes our measurements will separate maximally (1024→896, ratio 0.875, and 896→768, ratio 0.857) receive boundaries within 0.01 of each other at *every* δ.

3.5 Finite draws and the debiased estimator

The estimand of Eq. 1 is defined on population mean gradients; any instrument estimates them from D noise draws. This finite-draw step is not a neutral nuisance: in principle it biases the naive estimate in exactly the direction under test, so the correction belongs to the theory, not to the lab notebook.

Why finite draws manufacture floors. The natural estimate of Eq. 1, a redraw floor minus a single demoted-arm cosine, is asymmetric: the floor is a cosine between *two* finite-draw estimates of the native gradient, while each demote arm contributes *one* estimate compared against the native mean. Finite-draw noise attenuates every such cosine below its population value (each estimated direction carries variance the population direction lacks), and the attenuation is arm-dependent: per-draw gradient variance grows as the token count falls, because the MSE loss averages over fewer elements,

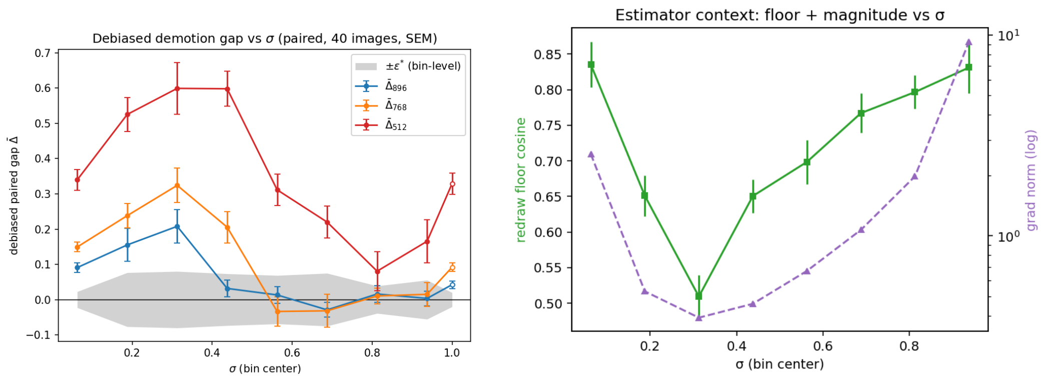


Figure 1: The verdict, before the apparatus. (a) Debiased demotion gap per σ-bin, paired against the re-encoding control (1024-tier natives, N=40; values in Table 1; gray band: bin-level $\pm\varepsilon^*$, Eq. 2; open markers: the σ=1 endpoint bin): safety is a per-route boundary (the 896 route enters the band at $\sigma \approx 0.5$, 768 has no certifiable window, and 512 does not enter at any noise level), against the spectral family's prediction of $\sigma \approx 0.14$–0.20 (Fig. 3, §4.2). (b) The estimator context those curves are read through: the redraw-floor cosine and the U-shaped gradient norm $\|\bar{g}_{\text{src}}(\sigma)\|$ (§4.4).

so a demoted arm's single estimate is noisier than the native estimates it is compared to. An iso-direction null (demotion changing *nothing* but the variance) therefore still produces a positive gap, and one that *grows as the target grid shrinks*: the same signature as "absolute token count sets the floor." A token-count-ordered floor can thus be manufactured entirely by the estimator, and no naive read can tell the two apart; this is why every verdict in this paper is stated in debiased units.

The debiased estimator. Two corrections, used together. First, *self-floors with attenuation correction*: every arm runs a second independent draw set g'_a, giving a per-arm self-cosine $\cos_{\text{self}} = \cos(g_a, g'_a)$, so that each arm has the same two-estimate structure the native floor has. The debiased cosine is the classical split-half disattenuation correction,

$$\hat{c} \;=\; \frac{\cos(\bar{g}_{\text{src}},\, \bar{g}_a)}{\sqrt{\cos_{\text{floor}} \cdot \cos_{\text{self},a}}}, \qquad \widehat{\text{gap}} \;=\; 1 - \hat{c}, \tag{8}$$

which cancels the finite-draw attenuation of numerator and denominator to first order. Second, *draw-count extrapolation*: the attenuation of an accumulated-draw cosine decays as $1/D$, so sweeping nested draw subsets and fitting $\text{gap}(D) = \text{gap}_\infty + c/D$ reads off the draw-limit gap gap_∞ directly. The two corrections are mutually checking: if the disattenuation removes the finite-draw component, debiased estimates must come out D-flat ($c \approx 0$) under the sweep, a prediction the instrument verifies (§4.1).

4 Experiments

The two accounts agree that gaps shrink with σ and order by severity; everything else is discriminating, and each subsection below states the prediction it discharges, against criteria frozen before the runs, together with its designated probe, several of which double as falsifiers of the account's own assumptions. The experiments proceed one term at a time: the instrument, its validation, and the debiased verdict map it certifies (§4.1); the spectral family scored against that map at the boundary level (§4.2); the graph floor (§4.3); the data branch and its governors (§4.4); the floor's decomposition (§4.5); and the account confronted with held-out routes (§4.6, Fig. 2).

4.1 The instrument and the debiased verdict map

What we will show. Figure 1 is the paper's central measurement, stated before the apparatus that produced it. Panel (a) is the debiased demotion gap per σ-bin (the per-route excess over the re-encoding control, in cosine units) for 1024-tier natives demoted to 896/768/512: the mildest route

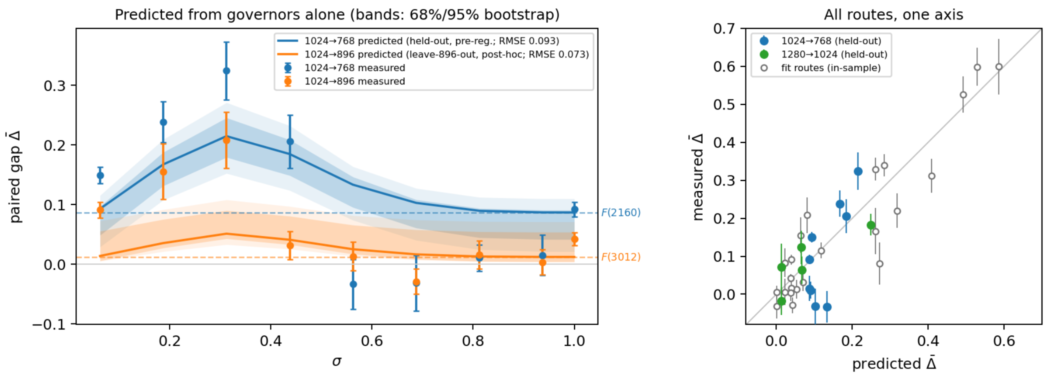


Figure 2: The account as a predictive theory (discharged in §4.6), under the headline exact angular link (Eq. 4; the three-form comparison is Appendix Fig. 4). Left: 1024-tier predictions with zero per-route freedom: the pre-registered held-out route 1024⟶768, and 1024⟶896 under the post-hoc leave-896-out re-derivation (so labeled; governors from 512+1120 alone). Dashed: each route's predicted floor $F(n)$; the gap between dashed and solid is the data term. Bands: 68%/95% full-pipeline bootstrap (resample images ⟶ re-bin ⟶ refit ⟶ re-derive governors ⟶ re-predict; B=1000; $\bar{m}$ and $\|\bar{g}_{\text{src}}\|$ are run-level instruments, held fixed). The 768 mid-σ dip sits outside the 95% band (§4.6). Right: predicted vs measured across all five routes and bins under the pre-registered pipeline; filled: the two held-out routes; open: fit routes (in-sample).

becomes safe only from $\sigma \approx 0.5$, the 768 route is not certified at any noise level, and the 2× downscale is gapped at every noise level, including pure-noise input. Panel (b) is the context every such number is read through: the redraw-floor cosine, which is what finite draws do to the estimator, and the U-shaped gradient norm $\|\bar{g}_{\text{src}}(\sigma)\|$ that shapes the curves (§4.4). The rest of this subsection builds the instrument behind the figure: the model and probe, the verdict table the figure plots, and the finite-draw ablation that forces every number in this paper into debiased units.

Model and data. All measurements use Anima (CircleStone Labs & Comfy Org, 2025), an open DiT-based flow-matching text-to-image model (2B parameters, 28 transformer blocks, 3D rotary position embeddings, and the Qwen-Image VAE, Wu et al., 2025), fine-tuned with LoRA adapters on an illustration corpus. Images are bucketed by native aspect ratio into resolution *tiers* indexed by nominal edge length $e \in \{512, 768, 896, 1024, 1280\}$; a tier-$e$ image occupies roughly $(e/16)^2$ latent-patch tokens (e.g. ~4,100 tokens at 1024, ~3,000 at 896, ~2,160 at 768, ~1,000 at 512). The probe adapter is a plain LoRA checkpoint trained at native tiers; we return to this operating-point dependence in §6.

The gradient probe. For each image and each of B uniform σ-bins we accumulate adapter gradients over D stratified noise draws per arm (verdict runs: 8×8 unless noted; $N = 40$ images) and compare arms by cosine similarity of the flattened accumulated gradients:

- **floor**: two independent draw sets at native resolution, the redraw null;
- **reenc**: decode–re-encode at native resolution, the control of §3.1;
- **demote arms**: one per edge e under test;
- **self-floors** (debiased runs): the per-arm second independent draw set the debiased estimator of §3.5 requires.

The gap is in cosine units (a mismatch *fraction*, scale-free; bin-mean SEM ~0.02 at N=40). Gap subtraction, not absolute cosines, is the valid read: the floor itself moves with $\|g\|$ across bins. Every bin-mean curve must pass a split-half reliability check over images, a discipline imposed after an earlier failure in which per-image rankings from the same gradients had split-half reliability indistinguishable from zero. Uniform σ-bins make the training-time σ-density irrelevant to the map; density-weighting the bins reproduces our earlier pooled measurements as a consistency check. Estimator details and the endpoint / x-zero probe modes are in Appendix C.

Table 1: **Debiased verdict map** (plotted in Fig. 1a): paired per-image debiased excess $\overline{\text{gap}}$ (SEM) against the re-encoding control, 1024-tier natives (N=40, D=8/bin plus a σ=1 endpoint bin at the same draws; non-finite and $|\text{gap}_{e,i}| > 1.5$ per-image values trimmed, retaining 35–40 images per cell). Bin-level ε^* at this (N, D) is 0.03–0.08 (Eq. 2).

σ bin center	1024→896	1024→768	1024→512
0.06	+.091 (.013)	+.149 (.014)	+.340 (.029)
0.19	+.155 (.047)	+.238 (.034)	+.525 (.048)
0.31	+.208 (.047)	+.324 (.049)	+.599 (.073)
0.44	+.032 (.024)	+.206 (.045)	+.598 (.050)
0.56	+.013 (.024)	−.034 (.042)	+.312 (.045)
0.69	−.029 (.021)	−.032 (.047)	+.220 (.046)
0.81	+.016 (.024)	+.011 (.022)	+.081 (.056)
0.94	+.004 (.021)	+.015 (.034)	+.166 (.061)
1.0 (endpoint)	+.043 (.011)	+.092 (.012)	+.329 (.030)

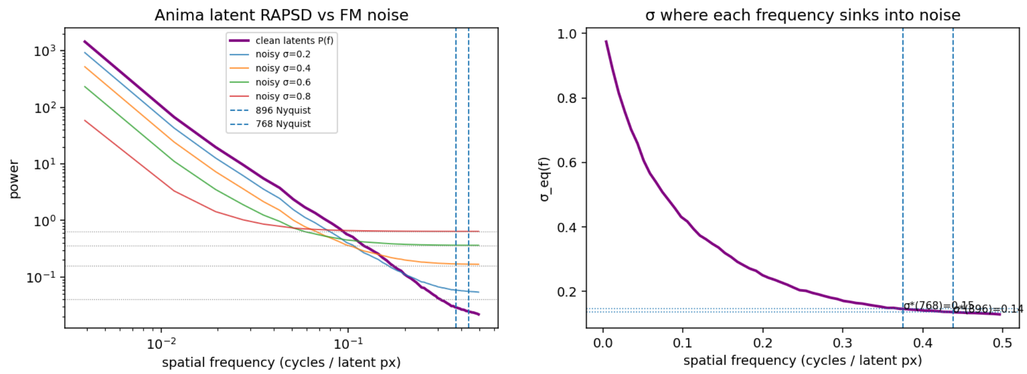
The measurement. Table 1 gives the debiased verdict map for 1024-tier natives demoted to 896/768/512, the numbers plotted in Fig. 1a (the uncorrected historical curves are Appendix Table 6).

Three reads, against criteria frozen before the runs: (i) σ-dependence is real and in the predicted direction, and tier ordering holds at every bin: the *qualitative* spectral intuition survives; (ii) the crossover for the mildest route is $\sigma \approx 0.5$ (the 896 route is clearly gapped through $\sigma \leq 0.31$, marginal at 0.44, and its means sit within ±0.03 of zero from 0.56 on), 3.5× the equal-power prediction; no tolerance rescues the spectral family, a boundary-level score deferred to §4.2 (Table 2); (iii) the 2× downscale route is *certifiably unsafe at every noise level*: its paired debiased lower bound stays above ε^* in every bin (means +0.08 to +0.60), including $+0.17 \pm 0.06$ at $\sigma = 0.94$, where the input is ~97% noise by power in every band the coarse grid cannot represent, a high-σ persistence that no member of the spectral family can produce. There is no noise level at which a 2×-downscaled latent yields the native gradient.

Two revisions against the historical record are part of the result. The 768 route's high-σ plateau was roughly half estimator bias: debiased means sit within noise of zero across $\sigma \in [0.56, 0.94]$, but with upper bounds that do not cross below the 0.02 margin at this (N, D), and a real endpoint floor of $+0.092 \pm 0.012$ remains; the earlier verdict of "unsafe at every noise level" accordingly softens to *no certifiable window at instrument resolution* (§5.1). Conversely the 896 route's endpoint shows a small *real* gap ($+0.042 \pm 0.011$) that single-estimate variance had buried; its safe window is the interior $\sigma \in (0.5, 0.94]$, not the endpoint itself.

Does finite-draw bias matter? The bias mechanism of §3.5 is live at our operating point, and large. A variance back-of-envelope predicts spurious endpoint gaps of $\approx 0.02/0.05/0.15$ for 896/768/512 (plausibly ~40% of the uncorrected intermediate floors), and a retroactive scan confirms the confound: the 1024→896 endpoint gap decays as +0.100/+0.035/−0.016 at $D = 4/8/16$, a clean c/D decay with fitted $c \approx 0.5$. The native floor itself makes the point sharpest: at D=8–16 the uncorrected endpoint self-cosine reads 0.79–0.85, but a nested draw sweep ($D = 4 \ldots 64$) extrapolates it to 1.005 (bootstrap 95% CI [0.994, 1.016]): the population-level per-bin native gradient is a *fixed direction*, and the entire redraw floor is finite-draw noise, not intrinsic gradient stochasticity. No conclusion in this paper survives on the naive estimator alone.

The debiased estimator, operated. Endpoint-mode runs sweep nested draw prefixes $D \in \{4, 8, 16, 32, 64\}$ (the D=64 pass contains the smaller draw sets, so one set of forwards suffices) and fit $\text{gap}(D) = \text{gap}_\infty + c/D$ per route with a bootstrap CI over images; verdict-grid runs compute Eq. 8 per image and bin. The mutual check of §3.5 passes: debiased fits are D-flat ($|c| \leq 0.05$ for the 512 route, against $c \approx +0.29$ uncorrected), so the attenuation correction removes exactly the $1/D$ component the sweep isolates. One caveat delimits where Eq. 8 is readable. At D=8 per bin, the *unpaired* debiased estimator is unstable in bins where the floors themselves are small (mid-σ; the denominator is noisy): there the readable object is the *paired per-image excess* $\text{gap}_{e,i} = \widehat{\text{gap}}_{e,i} - \widehat{\text{gap}}_{\text{reenc},i}$ of §3.1, with non-finite and $|\text{gap}_{e,i}| > 1.5$ values trimmed; pairing cancels the shared per-image,



(a) Latent RAPSD against the flow-matching noise floor at four noise levels; dashed lines mark the demoted grids' Nyquist cuts.

(b) The equal-power slice $\sigma_{\text{eq}}(f)$ of the spectral family. Predicted σ^*: 0.136 / 0.146 / ~0.20 for demotion to 896 / 768 / 512.

Figure 3: The spectral account, computed from the measured RAPSD and transported to the training question (§3.4): safety would be predicted from $\sigma \approx 0.14$. Against the measurement (Fig. 1) no tolerance reproduces the pattern (Table 2): the equal-power crossover is off by ~3.5×, and the 2× downscale, predicted safe by every tolerance, is measured unsafe at every noise level.

per-bin draw structure. All verdict maps below report this paired object. Finite-draw cosines are also compiler-kernel-path sensitive, so all pairings live within one run (Appendix C).

Measured resolution. At the verdict grid's operating point (N=40, D=8/bin) the bin-level ε^* (Eq. 2) is 0.03–0.08; endpoint-mode draw sweeps (D up to 64) tighten it to $\approx$0.01–0.02.

Units and coverage, stated once. Every number in the main text is debiased; the uncorrected historical tables are in Appendix E and are not cited as evidence. Debiasing is not a formality; it moved verdicts in both directions: it *raised* the mid-σ gaps the attenuation had hidden, halved the 768 endpoint floor, and surfaced a small real 896 endpoint gap that single-estimate variance had buried (revision record: Appendix Table 7). Four secondary probes (the 1280-tier iso-severity twin, the 896-tier transfer, the depth split, and the positional-interpolation intervention) predate the debiased instrument; their reads are retained only where one of two defenses applies: the finite-draw bias is strictly positive, so an in-band read is conservative; and paired same-grid contrasts cancel the bias to first order. Their tables are likewise in Appendix E.

4.2 The spectral account, confronted at the boundary level

The spectrum. The Qwen-Image VAE's latents are high-frequency-quiet ($P(f) < 1$ above $f \approx$ 0.16 cycles/latent-pixel; latent variance 0.42), giving the equal-power crossovers $\sigma^*(896) = 0.136$, $\sigma^*(768) = 0.146$, $\sigma^*(512) \approx 0.20$, with tight per-image spread (p10–p90 $\approx$ 0.11–0.16): the spectral crossover is image-generic, and at that tolerance demotion would be safe over almost the entire σ axis (Fig. 3).

The tolerance family, scored at the boundary level. Table 2 instantiates Eq. 7 on the measured latent spectrum (P at the 896/768/512 cuts = 0.025/0.029/0.065). Because the latents are high-frequency-quiet, P barely varies across the cuts and the family is compressed: the spread between the mildest and harshest route remains below ~0.13 in σ at every tolerance. No row reproduces the measured pattern of §4.1: the δ that matches the measured 896 boundary (δ = 0.025) predicts 768 safe at 0.52 and 512 safe at 0.62, while SPD's own published δ = 0.01 predicts all three routes safe by 0.72, and no δ anywhere in the family can produce a σ-independent floor.

Which tolerance? Eq. 7 needs a δ, and it is worth asking where one could legitimately come from. SPD's published δ = 0.01 is an inference-side choice, selected on a speed–quality Pareto ablation over generated images (Xiao et al., 2026); nothing ties it to gradients. Machine precision

Table 2: The spectral account (Eq. 7) instantiated on the measured latent spectrum: predicted safe boundaries t^* per route as the tolerance δ sweeps its range. SPD's published default is $\delta = 0.01$; the equal-power slice is the classical crossover; δ_{reenc} is the measured re-encode noise floor, the zero-free-parameter anchor fixed by our own pipeline. No row reproduces the measured pattern (last line; §5.1): the δ that matches the safe route's boundary predicts both failing routes safe by $\sigma = 0.63$, and no δ can produce a floor.

tolerance δ	1024→896	1024→768	1024→512	
$(1 + P_\omega)/2$ (equal power)	0.14	0.15	0.20	classical crossover
0.05	0.41	0.43	0.53	
0.025	0.50	0.52	0.62	tuned to the measured 896 gate
0.01	0.61	0.63	0.72	SPD's published default
0.005	0.69	0.71	0.79	
$\delta_{\mathrm{reenc}} \approx 10^{-5}$	0.98	0.98	0.99	the parameter-free anchor
measured, debiased	≈ 0.5	none certified	unsafe, all σ	floors .02–.04 / .06–.09 / $\approx$ .3

Table 3: The floor, debiased. **Endpoint**: at σ=1 the input is exactly ϵ on every grid, so the input-mediated data term vanishes by construction. **x-zero**: the image is removed from input *and* target; any surviving gap is pure graph-shape sensitivity. Debiased draw-limit values $\widehat{\mathrm{gap}}_\infty$ [bootstrap 95% CI over images] from nested draw sweeps (endpoint: N=12, $D = 4 \ldots 64$; x-zero: N=40, $D = 4 \ldots 32$). The endpoint and x-zero columns agree within CI at every route, and the account's pre-registered kill switch (all endpoint gaps ≈ 0) does not fire.

route	endpoint $\widehat{\mathrm{gap}}_\infty$ [95% CI]	x-zero $\widehat{\mathrm{gap}}_\infty$ [95% CI]
native self-floor (cosine)	1.005 [0.994, 1.016]	—
re-encoding control	−0.003 [−0.017, +0.008]	—
1024→896	+0.019 [+0.010, +0.030]	+0.034 [+0.017, +0.058]
1024→768	+0.056 [+0.043, +0.071]	+0.074 [+0.053, +0.094]
1024→512	+0.304 [+0.197, +0.424]	+0.283 [+0.232, +0.332]

is not the relevant floor either: bf16 rounding on unit-scale per-band amplitudes injects power of order 10^{-5} or below, orders of magnitude beneath any boundary-moving tolerance. Our setting, however, fixes a tolerance *by construction*: the demotion recipe pays a resize–VAE-encode round trip whose latent error has measurable per-band power $D(f)$ (the same pipeline noise the re-encoding control prices at the gradient level), and D is expressed in exactly Eq. 7's units (per-band power against unit noise). Band content below $D(f_{\mathrm{cut}})$ is invisible to the substitution pipeline even with no resolution change, so $\delta_{\mathrm{reenc}}(e) := D(f_{\mathrm{cut}}(e))$ anchors the family with zero free parameters. Measured on the probe set (N=40, same resize–encode chain as the re-encoding control), the anchor is tiny and image-generic: $\delta_{\mathrm{reenc}} \approx 1.0 \times 10^{-5}$ at all three cuts (p10–p90 within $[1, 2] \times 10^{-5}$; per-band SNR $P(f_{\mathrm{cut}})/D(f_{\mathrm{cut}}) \approx 2.5$–$4.4 \times 10^3$), pushing the anchored boundaries to $t^* \approx 0.98$ on every route (Table 2). The parameter-free anchor is therefore maximally conservative and still structurally wrong in both directions at once: it moves all three boundaries together (spread < 0.01), it certifies the measured-safe route only above $\sigma \approx 0.98$ where the measurement certifies it from $\sigma \approx 0.5$, and, like every member of the family, it predicts that the 2× downscale becomes safe, when no noise level does. Either way the confrontation does not hinge on the choice: the structural failures — ratio-only boundaries, and a finite safe boundary for every route — are δ-independent.

4.3 The floor is graph: three probes, one number

Three designated probes establish the graph term, and with it the premises of the reduction. At the $\sigma = 1$ *endpoint* the input is pure ϵ, identical in distribution across arms, so the input-mediated data term is zero by construction; per §3.2, the target-mediated share need *not* vanish there, so the endpoint alone does not isolate the graph term, and the next two probes discharge that step empirically. The debiased endpoint floors are +0.019/+0.056/+0.304 for 896/768/512 (Table 3); the 512 value clears the pre-registered confirmation bar ($\widehat{\mathrm{gap}}_\infty \geq 0.15$) on 12 of 12 probe images individually, so the token-count floor is not an estimator artifact.

The *x-zero* probe removes the image from input *and* target everywhere, isolating pure graph-shape sensitivity, and its debiased draw-limit gaps are statistically equal to the endpoint gaps at *every* route (Table 3). The *target-strength sweep* closes the remaining loophole from the α side: re-running the endpoint with target $\epsilon-\alpha x$, $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$ (N=12, D=16, one run, shared draws across α), the paired debiased excess is α-flat at the well-conditioned anchors (768: +0.070±0.015 at α=0 versus +0.049±0.010 at α=1; 512: +0.269±0.041 versus +0.337±0.067; re-encoding control slope +0.003): no resolvable target-content share *in the angular estimand*. The interior points $\alpha \in [0.5, 0.75]$ are unreadable by construction and excluded: there the native residual $\alpha x - \hat{x}$ passes near cancellation, the gradient norm dips (60 → 33) and the redraw floor falls to 0.87, so the estimator reads through a small gradient norm, the low-norm amplification of §4.4 surfacing along the α axis. Endpoint ≡ x-zero ≡ α-flat: three independent probes converge on one number per route.

Why the target term is angularly invisible: it lands parallel. The α-flatness is not a cancellation mystery, and it does not mean the target-mediated share of §3.2 is absent. Because the objective is quadratic, $\bar{g}$ is *affine* in α, so the exact target-mediated gradient $t = \bar{g}(1) - \bar{g}(0)$ is computable from the endpoint arms with zero draw noise beyond theirs; measuring it directly (N=40 endpoint, two independent draw sets) shows the term is real and large — $\|t_{\mathrm{src}}\|/\|\bar{g}_{\mathrm{src}}\| \approx 2.2$, so $J^{\top}$ does *not* annihilate target content — but demotion's change to it, $\delta t = t_{\mathrm{dem}} - t_{\mathrm{src}}$, lands almost entirely *along* $\hat{g}_{\mathrm{src}}$: $\kappa_{\parallel} = -0.75/-1.18/-1.86$ on 896/768/512 against $\kappa_{\perp} = 0.09/0.14/0.20$, an 8–9× parallel dominance, reproducible across the independent draw sets (δt direction agreement ≥ 0.995) while the re-encoding control sits at the noise floor with an irreproducible δt direction. Demotion *shortens* the target-mediated gradient along $\hat{g}_{\mathrm{src}}$, severity-ordered ($\|t\|/\|\bar{g}_{\mathrm{src}}\|$: 2.2 → 1.5 → 1.1 → 0.5); direction rotation appears only on the harshest route ($\cos(t_{512}, t_{\mathrm{src}}) = 0.74$ against 0.99/0.98 for 896/768). The cosine gap is blind to a parallel rescaling (Eq. 4), and the orthogonal part enters only at second order, $\kappa_{\perp}^2/2 \approx 0.004$–$0.02$, below the floors it would have to move: the α-flat scalar result stands *because* the target term lands parallel, not because it vanishes. (Per image the orthogonal loading is ~10× the aggregate's; image-specific directions cancel in the mean, the same motif as the residual-direction read of §4.4.) The endpoint gap *is* the graph floor as the angular estimand reads it, an empirical result established at our operating point, not assumed by construction.

How does the floor relate to the interior high-σ bins? For 512 the debiased window means (+0.08–0.60) sit at or above the floor, as additivity requires. For 896 and 768 the floors (0.02–0.04 and 0.06–0.09) are *at or below* the interior bins' certification resolution (ε^{*} = 0.03–0.08 at N=40, D=8): the endpoint sweeps resolve them only because their effective draw budget is an order of magnitude larger. The claim "the high-σ plateau is the floor" therefore sharpens to: the floor is what the plateau converges to when the data term dies and the estimator is given enough draws to see it.

Two further exonerations. The re-encoding control sits at zero within CI (debiased −0.003 [−0.017, +0.008]), so the VAE round trip is not the cause; conversely, the round trip has *measurable* input-level error while its gradient cost is below instrument resolution, so input-level error neither implies nor excuses gradient-level cost. And a forward-only probe of the model's caption-conditioned *prior* across grids shows route distances that are flat where the floors are strongly ordered, so the floor's route-ordering lives in the Jacobian factor J, not in the model's resolution-conditioned prior, and not in a training-distribution discontinuity (the 1280 grid, absent from training, is no more distant from 1024 than heavily trained adjacent pairs are from each other).

4.4 The data branch: a route-uniform numerator over a U-shaped denominator

$m(\sigma)$ is route-uniform in amplitude: the residual-level spectral prediction fails. Measuring the mean prediction residual $\bar{r} = \mathbb{E}_{\epsilon}[\hat{v} - (\epsilon - x)]$ per grid and σ (the exact r in $g = J^{\top} r$) shows a strong, monotone σ-shape (cross-grid excess 0.89 → 0.36 in relative L_2 from σ=0.125 to 1) that is the *same for every route* within ±0.02, across routes whose gradient floors span 0 to 0.3 and whose crossovers span 0.5 to nonexistent (Appendix Table 10). This kills the residual-level prediction of §3.4: with no cross-frequency coupling, its mismatch is confined to the destroyed band and ordered by destroyed-band energy, which differs substantially across these routes; route-uniformity says the real mismatch is carried by cross-band structure the diagonal model cannot express. All route identity therefore lives in J, not in the prediction error, a pre-registered contrast that would equally have falsified the account's factorization had the residual curves been route-ordered; instead it *establishes* the shape factorization directly. The decay is smooth, Wiener-like posterior shrinkage with no hard gate at the spectral crossover: gap curves read as "collapsed" where the data term sinks below the instrument's certification resolution, not at σ_{eq}.

A direction-resolved follow-up sharpens what "route-uniform" means: the uniformity is a property of the *amplitude* only. Saving the per-image mismatch vectors and comparing $\Delta\bar{r}$ directions with a split-half attenuation correction, non-adjacent route pairs are near-orthogonal at low σ (corrected cosine 0.00–0.08 at $\sigma \leq 0.375$) with only a weak shared component at high σ (+0.2–0.3 at $\sigma \geq 0.875$); the top mode of a stacked SVD carries 0.33–0.36 of the energy against the 0.25 rank-4-uniform baseline (a rank-one common mode would give ≈ 1); and cross-image direction consistency is zero at every (route, σ) cell (max +0.019 at split-half reliabilities 0.73–0.99), so the mismatch direction is fully image-specific — refuting a grid-conditional composition prior ("small canvas $\Rightarrow$ portrait framing") as the carrier, at least under full captions. What *is* shared is scale and spectral drift: $\Delta\bar{r}$ energy migrates toward low frequencies as σ rises (low-third share 0.40 $\rightarrow$ 0.67 by $\sigma = 0.875$), composition-scale but per-image. The factorization of Eq. 6 only ever consumes the norm $m(\sigma) = \|\Delta\bar{r}(\sigma)\|$, so the account is untouched; what sharpens is the open question, now "why is only the *amplitude* universal when the directions are image- and route-specific." A closed-form candidate for cross-band coupling per se exists — the flow posterior covariance, whose off-diagonal $D_z v^*$ entries are $-\big((1-\sigma)/\sigma^3\big)\,\mathrm{Cov}(x_\omega, x_{\omega'} \mid z)$ (Xing et al., 2026), set to zero by the diagonal-Gaussian premise — but its rank-one common-mode version is excluded by the SVD read above.

$\|\bar{g}_{\text{src}}(\sigma)\|$ shapes the curve. Because the gap is a cosine it measures a mismatch *fraction*; the total gradient norm is U-shaped in σ (2.6 $\rightarrow$ 0.4 at $\sigma \approx 0.3$ $\rightarrow$ 9.3; Fig. 1b): large at high σ where the model is pulling composition out of the prior and the residual is image-scale, small in the mid-σ refinement regime, and inflated again at low σ by irreducible ϵ. A monotone absolute mismatch divided by a U-shaped norm yields exactly the measured interior-peaked gap curves: bin-mean gap tracks $1/\|\bar{g}_{\text{src}}\|$ (Spearman +0.55 to +0.90 across arms and runs, and within images), and renormalizing by $\|\bar{g}_{\text{src}}\|$ flips the anomalous low-σ dip into the monotone maximum in every arm of both runs, the predicted $m/\|\bar{g}_{\text{src}}\|$ shape. This is a post-hoc consistency analysis (so labeled): it explains the *shape*; all safety criteria stay in cosine units, since direction is what a normalized optimizer consumes.

The governors: ratio sets the amplitude, absolute size sets the floor. With the floor isolated, what orders the routes? Two probes give a clean division of labor. *Absolute size, not ratio, sets the floor*: the route 1280→1024 is *more* aggressive by ratio (0.80) than 896→768 (0.857, no certifiable window), yet its floor is ≈ 0 and it enters the re-encoding band at $\sigma \gtrsim 0.75$, refuting any pure ratio rule for the floor, and with it the spectral account's only degree of freedom. (The 896→768 read is direct: re-run one tier down, the route fails its pre-registered bar with a high-σ residual ~2× the 1024→896 plateau, at ratio 0.857 against the passing 0.875, while the spectral family puts these two boundaries within 0.01 of each other at every tolerance; Appendix Table 9.) Conversely the floor grows monotonically as the *target* grid shrinks (0.02–0.04 / 0.06–0.09 / $\approx$ 0.30 at target ~3,000/2,160/1,000 tokens), consistent with "floor = how well the coarse graph approximates the fine graph's computation." *Ratio, not absolute size, sets the amplitude*: a synthetic iso-severity route 1280→1120 (edge ratio 0.875 exactly matched to 1024→896, but 1.6× its absolute target size) reproduces the 1024→896 curve bin-for-bin and its crossover $\sigma^* \approx 0.5$, while the same-native 1280→1024 (ratio 0.80) separates and floors later (Appendix Table 8). Amplitude follows ratio; with $\text{Floor}_{1120} \approx 0$ this completes the division: *ratio sets A_e; absolute target capacity sets* Floor_e. The pre-registered contrast (ratio-governed $\Rightarrow \sigma^* \approx 0.5$; capacity-governed $\Rightarrow$ earlier floor) discriminated cleanly. Note what survives of the spectral account here: ratio *is* the right governor for the data term; its error is not its governor but its scope, mistaking one term for the whole gap.

4.5 Decomposing the floor: depth, and a positional share

Depth, not module type. Splitting the probe gradients per module type (15 groups, including separate rows of the fused QKV projections) and per block (28) shows the floor concentrated in *depth*: early blocks (0–9, peaking at 3–8) carry ~3× the late-block gap, uniformly across every module type within a block (at $512/\sigma{=}1$, every type sits in 0.22–0.31; Appendix Table 11). The mechanism picture: the first ~10 blocks build grid-calibrated token statistics; the divergence propagates into every parameter type's gradient in those blocks roughly equally, and deeper blocks inherit a washed-out version. Notably, query and key projections show *zero* excess over value projections, which we initially read as refuting a rotary-embedding mechanism. That inference was wrong, in an instructive way.

$\mathrm{Floor}_e = \mathrm{RoPE}_e + \mathrm{Resid}_e$. Landing-side uniformity cannot localize *origin*: a perturbation originating in the position embedding propagates through the block and lands on all module types. The discriminating experiment is the origin-side intervention of §3.2: evaluate the demoted grid with positionally-interpolated fractional rotary coordinates (patch i at $i \cdot H_{\mathrm{src}}/H_{\mathrm{dem}}$ per axis), making its relative phase geometry exactly match native. At the σ=1 endpoint this *erases* the 768 floor (paired $\Delta = +0.081$, 78% of images improved) and removes ∼30% of the 512 floor (paired $\Delta = +0.096$), while leaving the in-band 896 control untouched (Appendix Table 12; the paired Δs are same-grid contrasts at matched draws, in which the finite-draw bias cancels to first order). Re-anchored against the debiased floors: the +0.081 erasure accounts for essentially all of the 768 floor ($\mathrm{Resid}_{768} \approx 0$); against the 512 floor of ≈ 0.30, the +0.096 erasure leaves a $\mathrm{Resid}_{512} \approx 0.2$ bulk. The mild-route floor is mostly a *coordinate-system artifact*; the harsh-route bulk is a genuine non-positional graph residue (attention-over-N statistics, sequence-length-dependent normalization, capacity). The capacity governor of §4.4 attaches to Resid_e, as the band-counting sketch anticipated. The attention-over-N piece has inference-side prior art: attention-entropy corrections derive length-dependent temperatures to preserve the $\log N$ entropy term (Li et al., 2025), and TIDE applies the same dilution correction jointly to resolution and diffusion timestep (Liu et al., 2026) — the closest prior work to a σ-gated resolution correction, though on the inference side; our gate is training-side and set by measured gradient safety, not by an entropy invariance. The 896 route's small floor (0.02–0.04) sits below the PI probe's paired resolution and remains undecomposed.

The floor as a ledger. Combining the designated probes, the floor decomposes additively, each share measured by its own intervention:

$$\mathrm{Floor}_e = \underbrace{\text{reenc}}_{\approx 0 \text{ by control}} + \underbrace{\text{target content}}_{\text{lands } \parallel \hat{g}_{\mathrm{src}}\text{: angular share} \approx 0} + \underbrace{\mathrm{RoPE}_e}_{\text{erased by PI at } \sigma=1} + \underbrace{\mathrm{Resid}_e}_{\text{remainder}}$$

with totals 0.02–0.04 (896; undecomposed), ≈ 0.07 (768; RoPE the large majority, Resid ≈ 0), and ≈ 0.30 (512; RoPE ≈ 0.10, Resid ≈ 0.20). One caveat on the first share: the re-encoding control (native decode→re-encode) is a *proxy* for the pipeline cost demotion actually pays (source downscale→encode); the demote–re-promote arm (§4.6) closes that gap empirically. The data leg's orthogonal magnitude $|B^{\perp}|$ measured against the re-encoded reference agrees with the native-referenced read to ∼4% at the signal-carrying low-σ bins (worst ±23% only in bins where B is already small), so the shared downscale→encode pipeline cost is a minor share of the data intervention (Appendix Table 5).

The positional share is a noise-regime artifact only. A σ-resolved follow-up closes the tempting route this opens (training 1024→768 through interpolated coordinates): with image content in the input, the stretched forward is *off-manifold*; the interpolated arm is *worse* than the plain demoted arm through $\sigma \in [0.56, 0.81]$ (paired −0.05 to −0.11) and wins only at $\sigma \geq 0.94$. The floor's positional share is erasable only where the input is noise-dominated, so the decomposition is a mechanism finding, not a training lever; the data term of the 768 route is in any case fatal on its own (+0.09–0.22 over control across the window). A frequency-*selective* banded alignment does survive in-window (§5.2).

4.6 The account confronted: held-out routes and the reduction's domain

The probes so far test the account's *structure*. The remaining test is whether it is a *predictive* theory: fit the reduced account on three routes, then predict two held-out routes from the governors alone (Fig. 2). Fit routes: {1024→896, 1024→512, 1280→1120}; held-out: {1024→768, 1280→1024}. Per fit route the excess curve $\overline{\mathrm{gap}}_e(\sigma)$ is fit with a route amplitude and floor on the measured route-uniform amplitude $m(\sigma)$ and each run's own $\|\bar{g}_{\mathrm{src}}(\sigma)\|$; two governor models ($A$(ratio) interpolating the fitted amplitudes, and an exponential floor law $F(n) = F_0\, e^{-n/\tau}$ in target tokens n) then generate the held-out predictions with zero per-route freedom. Pass criteria were pre-registered in the analysis script before the numbers.

Held-out results. All four pre-registered gates fire: (i) held-out 1024→768: RMSE 0.093, *better than an oracle quadratic fit on the held-out data itself* (0.105), and better than the spectral account on its committed region ($\sigma > \sigma_{eq}$, where it predicts gap 0: RMSE 0.164 vs 0.096 ours); (ii) held-out 1280→1024: RMSE 0.092 vs oracle 0.056, inside the pre-registered 2× gate; (iii) the ratio governor holds at the amplitude level: the two ratio-0.875 routes agree at $z = 0.14$ despite a 1.6× difference in target capacity; (iv) the floor law $F(n) = 0.70\, e^{-n/1041\,\mathrm{tok}}$, fit on the 512 and 896 floors, predicts $F(4825) = +0.007$ for the 1120 grid (fitted +0.002) and $F(2160) = +0.088$ for the held-out 768 grid, against a measured endpoint floor of +0.092 ± 0.012. The point estimate lands exactly, but its own sampling spread is wide: a full-pipeline bootstrap (Fig. 2; §5.2) puts $F(2160)$ at 68% $[+0.042, +0.091]$, consistent at the interval's upper edge, so the licensed read is agreement, not exactness. A post-hoc leave-896-out addendum (so labeled) re-derives the governors from 512+1120 alone and predicts the 896 floor at +0.019 against the measured +0.019 $[+0.010, +0.030]$: the floor law is correct on both floors outside its fit set.

The floor law is a calibrated interpolation, not a mechanism. Two anchors and one held-out check cannot identify a functional form, and the cosine gap is the wrong scale to fit one in: it saturates (Eq. 4). Refitting the same anchors in the unsaturated perturbation-energy units, $\kappa_{\mathrm{eff}}^2 = (1-F)^{-2} - 1$, gives $\tau \approx 860$ tokens and predicts $F(2160) = +0.096$, equally consistent with the measured +0.092 ± 0.012; and the same two anchors pushed through $e^{-\sqrt{n}/\ell}$ or a power law n^{-p} predict +0.085 and +0.075 — every form lands within the union of the measured CI and the prediction's own bootstrap spread (§5.2). The exponential is therefore one member of a candidate family $\{e^{-n/\tau},\ e^{-\sqrt{n}/\ell},\ n^{-p}\}$ our operating points cannot distinguish, and the mechanical reading of τ ($\approx$ 860–1040 tokens across the two unit conventions) as an effective-rank decay length of the early-block gradient operator is a hypothesis, not a law. It is a discriminable one: a spectral-tail account predicts a source-capacity dependence $F \propto e^{-n/\tau} - e^{-n_0/\tau}$ where the absolute-target-capacity governor of §4.4 predicts none (at our operating points the secant correction sits within calibration slack, so the existing anchors do not discriminate); the designated run is a fixed-target, varied-source token ladder, not yet scheduled.

Functional form: the exact link headlines. Refitting the same pipeline under three forms for the data term (a fitted power $A\, m/G^p$ with p shared and free, the derived quadratic $A\,(m/G)^2$, and the exact angular link, Eq. 4) resolves the form question in the geometry's favor twice over (Appendix Fig. 4; on the fit routes the three forms are visually indistinguishable, so Fig. 2 carries the headline form only). First, the free exponent's own optimum lands on $p = 2.00$ exactly: the "empirical" power form recovers the derived quadratic on its own. Second, the exact link is the best held-out predictor (mean RMSE 0.071 vs 0.093–0.094 for the other two), and its entire margin comes from the 1280→1024 route (RMSE 0.049 vs ≈ 0.095; Appendix Fig. 4): the quadratic systematically *overshoots* that route's mid-σ peak, and the exact link's saturation removes the overshoot; what a first analysis had booked as convexity in the $A(\mathrm{ratio})$ governor was quadratic-form error. The governors are form-invariant (floor law $F(2160) = +0.087$–0.088 under all three; ratio-twin $z \leq 0.44$). Per the ledger of §3.3: Eq. 4 is the best empirical predictor (empirical through its linear loading $c_e\, m$), and the derived quadratic is its small-κ limit: one geometry at two orders, not two competing models.

Stated limits. The prediction is *not* within certification resolution anywhere (χ^2/bin 7.6–9.2 held-out under the quadratic): the licensed claim is shape, magnitude class, and governors at ~0.07–0.09 RMSE, not "predicts within ε^*." And the $A(\mathrm{ratio})$ interpolation rests on two distinct ratio values.

The reduction's domain: one signed failure, and its signature. The 768 route's mid-σ window is the reduction's one structural failure, and its shape is diagnostic. Measured excess sits at zero across $\sigma \in [0.56, 0.94]$, *below* the predicted $\mathrm{Floor}_{768} \approx 0.09$ and outside the bootstrap 95% band (Fig. 2). This is structural, not sampling noise: a positive two-term form cannot produce a gap under its own floor, under any coefficients. In the four-term expansion (Eq. 5) a gap below the floor admits exactly two mechanisms, and we pre-registered both ahead of the direct probe: either (i) the projected interaction is negative in the window, $I_{768}(\sigma) < 0$ — in which case an amplitude-matching account predicts the window center sits where the two perturbation legs have equal orthogonal magnitude, a testable localization — or (ii) the graph share is itself σ-dependent, $F_{768}(\sigma)$ collapsing below its endpoint value in-window, which would rewrite assumption (iii) rather than (ii). Either way a named assumption fails exactly where the account says it can, and the derived four-term form represents the failure rather than excusing it. The window therefore *delimits the reduction's domain*. One honesty note on the probe design: the scalar read we originally planned, $\mathrm{gap}_{\mathrm{demote}} - \mathrm{gap}_{\mathrm{repromote}} - \mathrm{Floor}_e$, cannot arbitrate — it estimates $I(\sigma) + [F(\sigma) - F(1)]$ plus higher-order terms, conflating the two branches. The designated direct test is therefore vector-resolved: the demote–re-promote probe stores arm-mean gradients and reads the data, graph, and interaction shares per (route, σ-bin) from the interventional split $B = \bar{g}_{\mathrm{repromote}} - \bar{g}_{\mathrm{src}}$, $C = \bar{g}_{\mathrm{demote}} - \bar{g}_{\mathrm{repromote}}$ ($S = \|B^{\perp}\|^2/2\|\bar{g}\|^2$, $F = \|C^{\perp}\|^2/2\|\bar{g}\|^2$, $I = \langle B^{\perp}, C^{\perp}\rangle/\|\bar{g}\|^2$, with cross-draw-set inner products to cancel shared-noise bias and exact counterfactual angles alongside the quadratic ledger; instrument details and the full ledger in Appendix Table 5).

The probe, run: branch (i), negative interference. The data picks branch (i), with the pre-registered localization confirmed. $I_{768}(\sigma) < 0$ at every bin (-0.31 at $\sigma = 0.56$, shrinking in magnitude to -0.014 by $\sigma = 0.94$), with the two legs' orthogonal components near anti-parallel in-window ($\rho = \cos(B^{\perp}, C^{\perp}) \approx -0.93$; the sign is robust to the cross-set debiasing — same-set -0.39 vs cross-set -0.31 at the first bin). Branch (ii) did not fire: $F_{768}(\sigma)$ decreases monotonically to its endpoint value ($0.200 \to 0.004$) and never drops below it in-window, so the σ=1 endpoint reads of §4.3 are untouched. The amplitude-matching localization lands where branch (i) predicts it: $|B^{\perp}|/|C^{\perp}| = 0.98$ at the $\sigma = 0.688$ bin, putting the predicted window center at $\sigma \approx 0.69$, interior to the measured $[0.56, 0.94]$ window. The anti-parallel geometry itself is universal — $\rho \in [-1.24, -0.62]$ across every route and bin — so what distinguishes the routes is amplitude matching, not sign: 896 is matched but low-amplitude (net S+F+I $\leq +0.012$), 768 is matched at mid-window ($\leq +0.029$), and 512 is amplitude-mismatched at low σ ($|C^{\perp}|/|B^{\perp}| \approx 1.8$, net up to +0.20).

How much of the gap the interference erases. Because $\bar{g}_{\mathrm{dem}} = \bar{g}_{\mathrm{src}} + B + C$ by construction, the realized demotion distance is exactly the counterfactual angle $h(B{+}C)$, where $h(X) = 1 - \cos(\bar{g}_{\mathrm{src}}, \bar{g}_{\mathrm{src}} + X)$ is the distance the perturbation X alone would produce; $h(B)$, $h(C)$, and $h(B{+}C)$ are all computable from the stored arm means, so each account's predicted-to-realized ratio is direct. The no-interference scalar account S+F overpredicts the realized in-window gap 2.4–3.8× at 768 (1.6–5.8× at 896, 1.2–3.0× at 512); the fully additive counterfactual $h(B) + h(C)$ overpredicts 3.5–8× — the realized $h(B{+}C)$ is only ~20–30% of the additive sum in-window, i.e. roughly three quarters of the additively-predicted gap is erased by the interference. That is the quantitative content of branch (i); the vector account B+C is exact by construction — including I is what removes the overprediction. One unit-honesty caveat: in-window $|B^{\perp}|, |C^{\perp}| \approx 0.5$–$1.0\,\|\bar{g}_{\mathrm{src}}\|$, outside the quadratic truncation's domain, and the truncated ledger S+F+I correspondingly *under*predicts realized magnitudes ~3–4× (median ratio 0.24×). The ledger therefore licenses sign, decomposition, and window localization; gap magnitudes are quoted from the exact counterfactuals $h(\cdot)$ throughout.

The spectral account, scored. Against our account's discriminating predictions, the spectral account loses each one it is party to: the boundary pattern is $\{\approx 0.5$, no certifiable window, certifiably unsafe everywhere$\}$ (§4.2); gaps persist at pure-noise input, equal to the floor and to the image-free control (§4.3); ratio governs the amplitude while absolute size governs the floor (§4.4); the residual curves are route-uniform in amplitude under route-ordered gaps (§4.4); and the curve shape follows the $m/\|\bar{g}_{\mathrm{src}}\|$ read, not per-band SNR. The account also retrodicts the observations that motivated no probe: (i) $\sigma^* \approx 0.5$ rather than 0.14: the crossover is where the data term sinks below certification resolution, a property of the ratio, so the 3.5× discrepancy is expected, not anomalous; (ii) the 896→768 failure (the 768-grid floor alone fills the certifiable margin, so no σ-gate can rescue it); (iii) 1280→1024 flooring *later* ($\sigma^* \approx 0.75$) but cleanly (harsher ratio ⇒ larger A_e; large target ⇒ Floor ≈ 0), the single observation that breaks both pure-ratio and pure-capacity route-ordering stories. A curve-level overlay, in which each published tolerance δ is converted to a predicted gap curve through the account's bridge (destroyed-band Bayes-residual mismatch, measured $\|\bar{g}_{\mathrm{src}}\|$, route gain calibrated on the one safe route), is **[pending: analysis over existing rows; δ_{reenc} for the anchored line is now measured, §4.2]**; unit honesty requires saying it will be *the spectral account read through our bridge*, since it emits residual units only.

A corollary worth stating plainly: the safety of 1024→896 is an *empirical smoothness statement about this network's function across nearby token counts*, not an information-theoretic statement about the data. There was no a priori reason to expect a universal ratio invariant.

5 The answer, in deployable form

5.1 The (route, σ) safety map

The answer to the title question at our operating point is not a corrected formula but a measured map at a stated resolution; the map needs no table, because the verdict quantity self-describes it: a route is safe exactly where its debiased excess over the re-encoding control is indistinguishable from zero at instrument resolution (Eq. 2). Measured, on the *per-example* object: 1024→896 (ratio 0.875, floor 0.02–0.04) is **safe** on the interior window $\sigma \in (0.5, 0.94]$; 1280→1024 (ratio 0.80, floor ≈ 0) is **safe** from $\sigma \gtrsim 0.75$; the synthetic iso-severity grid 1280→1120 (ratio 0.875, floor ≈ 0) is safe with $\sigma^* \approx 0.5$, as its ratio twin predicts; 1024→768 and 896→768 (floors 0.06–0.09 and 0.06–0.12) have *no certifiable window*; and any route to 512 (floor ≈ 0.30) is *certifiably unsafe at every* σ. The 1280-tier pair and 896→768 were measured only by the pre-debiasing instrument, a conservative read since the finite-draw bias is strictly positive (§4.1). Safety is a per-route boundary σ^*(route), jointly determined by ratio (through A_e) and absolute target size (through Floor_e); neither alone predicts the ordering. For the corpus safe route the fine print is: every bin in $(0.5, 0.94]$ has a debiased mean within ±0.03 of zero, formal certification at the strict 0.02 margin is achieved at $\sigma = 0.688$ (elsewhere bin-level ε^* exceeds it at the current N, D), and the σ=1 endpoint itself carries a small real gap (+0.042 ± 0.011): the safe window is the interior.

The map depends on the aggregation object. Per §3.1, the aggregation operator is part of the estimand, and the two objects genuinely disagree at high σ: pooled-arm probes (pool size 4) collapse the 1024→896 excess to ≈ 0 at every bin $\sigma \geq 0.625$ *and* the 1024→768 excess to ≈ 0 at $\sigma \geq 0.875$: the batch-aggregate object SGD follows is more forgiving here than the per-example read, and the small per-image residual behaves as draw noise that cancels across images rather than as a shared bias. We publish the per-example map as the verdict (pooled arms do not yet have self-floors), read under the decomposition of §3.1: the pooled collapse says the $1/B$ term dominates at these bins, but the intercept — the coherent drift no batch size removes — is bounded only by an $a + b/B$ fit across pool sizes, the designated headline readout of a debiased verdict grid at the shipped trainer's actual batch×accumulation operating point **[pending]**. A redundancy-stratified analysis finds no per-image demotion-targeting lever (null trend across strata), consistent with the floor being an image-generic network property.

5.2 σ-conditional training on the measured-safe route

We wire the one measured-safe corpus route into the LoRA trainer as an opt-in path, exactly as measured, with no extrapolation beyond the map:

σ-first draw. The trainer draws each batch's σ *before* fetching latents, from the unchanged training density (the σ marginal is untouched; demotion conditions on σ, not the reverse). When every sample in the batch is above the gate ($\sigma > 0.5$), the native 1024-tier latent is swapped for a cached demoted-grid sibling produced by the probe's own measured-safe recipe (pixel-space downscale, VAE re-encode); everything downstream (noise, target, masking, loss) derives from the swapped latent. Off-route images always train native, as does validation. The small real gap at the exact σ=1 endpoint (§4.3) does not bite here: it is a boundary point of measure zero for the continuous training σ-density, and the last interior bin ($\sigma \in (0.875, 1)$) reads ≈ 0 debiased.

Banded rotary alignment, σ-gated. Motivated by §4.5, a frequency-selective alignment (YaRN-style bands: full positional-interpolation stretch for rotary bands with few rotations across the demoted extent, native spacing for high-frequency bands, ramp between, with the band thresholds scheduled by a sigmoid gate in σ following Zhao et al., 2026) is applied on demoted forwards only. Unlike uniform interpolation it is *not* off-manifold in-window: it passed both pre-registered legs; it erases the static variant's low-σ liability (+0.064 → +0.033 ± 0.025) while preserving its high-σ gains (best and most stable arm at $\sigma \geq 0.59$ and the endpoint, paired −0.05 vs plain demotion). It is a refinement candidate, not a gate-widener: the gate stays $\sigma > 0.5$.

Cost accounting. With ~50% of draws above the gate and the demoted forward at ~0.72× tokens, the fixed-step epoch cost is $\approx 0.5 \cdot 0.72 + 0.5 = 0.86$, a *projected ceiling* of ~14% wall-clock on our corpus (96% of records are on the safe route), scaling with high-tier content. On the fixed-step exercise grid below, the realized saving is −15.1% forward FLOPs (exact operator-level counts over

Table 4: Sample-level footprint vs. the seed lottery: paired PE-Core cosine at matched (prompt, noise seed), mean ± seed spread (SD over the three matched-seed pairs; seed pairs for the yardstick row) over the frozen prompt set, on two single-artist corpora (60/15 training images). The yardstick row is the same cosine between native checkpoints that differ *only* in training seed; an arm reading at or above it perturbs endpoint samples less than a seed change does. **Bold**: the closest-to-native arm per corpus; the shipped path is the only arm at or inside the yardstick on both corpora.

arm pair (matched training seed)	corpus A	corpus B
native ↔ native, cross-seed (yardstick)	0.9558 ± .0046	0.9541 ± .0093
shipped (1024→896, σ>0.5 + alignment) ↔ native	**0.9551 ± .0057**	**0.9641 ± .0023**
gate off (unconditional 1024→896) ↔ native	0.9504 ± .0179	0.9500 ± .0141
off-map route (1024→768, same gate + alignment) ↔ native	0.9503 ± .0013	0.9553 ± .0127
negative control (unconditional 1024→768) ↔ native	0.9527 ± .0118	0.9545 ± .0127

each run's realized token histogram) and −14.6% wall-clock (n=3 seed means on two single-artist corpora; wall tracks FLOPs ≈ 1:1, cache-load overheads included): the projection is achieved. The claim is wall-clock at fixed steps, not "more steps in the same time."

Weight-space sanity. Under bit-exact deterministic training (paired runs; without determinism, twin runs put a chaos floor of 0.41 on checkpoint-displacement cosine, which would swamp the read), the rotary refinement adds no displacement over the demotion arm it refines (base↔refined 0.319 vs base↔demoted 0.320), with its footprint concentrated in the low-signal early/mid blocks, consistent with §4.5's depth profile.

Sample-level footprint. A fixed-step exercise grid (5 arms × 3 training seeds × 2 single-artist corpora, 480 optimizer steps each, bit-exact deterministic paired-RNG so every arm sees identical draws) renders a frozen prompt set at matched (prompt, noise-seed) pairs per checkpoint and embeds each render with a frozen vision encoder (PE-Core, pooled and unit-normalized; Bolya et al., 2025). Table 4 reads each arm's paired cosine against its native twin, judged against a seed-noise yardstick: the same cosine between native checkpoints differing only in training seed. The shipped path is the only arm at or inside the yardstick on both corpora (0.9551 vs 0.9558, a tie at instrument resolution; 0.9641 vs 0.9541): swapping every $\sigma > 0.5$ step to the demoted sibling perturbs endpoint samples about as much as changing the training seed. All three controls (the gate removed, the off-map 1024→768 route under the same gate and alignment, and the unconditional off-map negative control) fall below the yardstick on at least one corpus, an ordering consistent with the map; but the margins are small relative to the seed spreads in Table 4, so this instrument certifies the absence of a gross sample-level footprint, not quality non-inferiority; that read belongs to the CMMD gate below. Arm orderings shuffle between seeds; single-seed visual impressions of the grid are lottery reads.

The account's predictions, under the same discipline. The yardstick's lesson, that point estimates ride the draw, applies equally to the predictive account of §4.6 standing behind the map. A full-pipeline bootstrap (resample the verdict runs' images, re-bin, refit every route, re-derive both governors, re-predict; B=1000, fixed seed; bands in Fig. 2) licenses two reads. The 768 mid-σ failure sits outside the 95% band (structural, not sampling chance), while the floor law's headline hit softens from exact agreement to consistency: $F(2160)$ bootstraps to 68% [+0.042, +0.091] against the measured +0.092 ± 0.012, the point estimate's exactness sitting at the interval's edge. The bootstrap also exposes the pipeline's own branch fragility: the floor law's positive-floor filter re-anchors its legs in 41% of draws (Appendix C for the tally). We deploy the *measured* map (§5.1), not the predicted one, for exactly this reason.

What this section does not claim. The generation-quality acceptance gate (CMMD, Jayasumana et al., 2024, non-inferiority at training scale against the realized wall-clock saving) remains open. The exercise grid's first scoring pass used rating-restricted display prompts against mismatched reference pools and could not separate even the off-map control from native, so we read no quality verdict from it; the pre-committed full-band rescoring closes the line (a regression closes it negatively). Throughput and sample-level footprint above are measured; quality non-inferiority is not. We flag this deliberately: the gap between a 480-step exercise grid and production-scale training is exactly the kind of gap this paper is about.

6 Limitations

One model family, one operating point. All gradient probes ran on one DiT (Anima) with one adapter checkpoint trained at native tiers. A mixed-resolution-trained adapter might equalize (or widen) its own gradients; probing such a checkpoint first is the designated bound on how far the map generalizes. The floor's dependence on the fine-tuning operating point through J is untested (the designated probe: fine-tune at the 1280 tier, re-probe 1280→1024). Relatedly, the probe pool's relationship to the adapter's own fine-tune set was not a controlled axis (2 of 40 verdict stems are in it); a controlled-adapter 2×2 factorial (two LoRAs trained on opposite style clusters under frozen stem-level membership manifests, probed on both clusters) measured this dependence rather than conceding it: the map's shape replicates on both checkpoints and the adapter×probe-style interaction is below instrument resolution, while the absolute redraw-floor level is checkpoint-dependent (Appendix D).

Gradient-level, not end-task. The map is built from accumulated per-step gradients; integration over an optimizer trajectory is addressed by the fixed-step exercise grid's weight-space and sample-level reads (§5.2), at 480 steps rather than production scale, and the generation-quality (CMMD) gate remains open.

Account resolution and domain. The account predicts held-out routes at ~0.07–0.09 RMSE, not within ε^* (§4.6); its $A(\text{ratio})$ governor rests on two distinct ratio values; the reduction's domain excludes the 768 mid-σ window, whose mechanism the vector-resolved probe has since measured (negative interference, §4.6) but which the two-term form still does not predict; the floor law's functional form is unidentified at our operating points (two anchors and one check; §4.6); and the band-counting sketch for $\text{RoPE}_e/\text{Resid}_e$ is calibrated, not derived.

Instrument resolution. "Safe" is one-sided non-inferiority at the measured certification resolution (§3.1): $\varepsilon^* = 0.03$–0.08 per bin at the verdict grid's (N, D), ≈ 0.01–0.02 at the endpoint sweeps. True gaps below ε^* are undetectable by construction, the σ^* boundaries are localized only to a bin, and every "no certifiable window" claim is (N, D)-relative: a larger campaign could certify (or refute) the 768 high-σ window whose debiased means already sit near zero.

Debiased coverage. The debiasing campaign covered the corpus verdict grid: the endpoint draw sweeps, the 8-bin map, x-zero, and the α-sweep. The 1280-tier iso-severity, 896-tier transfer, depth-split, and PI-intervention probes remain pre-debiasing reads, defended where used by the one-signed-bias argument and by paired same-grid contrasts (§4.1); pooled arms have no self-floors yet. The 1280-tier curves enter the held-out validation with this caveat recorded.

Scope. Adapter (LoRA) gradients on a frozen backbone; full fine-tuning may differ. Pixel-space demotion only (latent-space downsampling was not tested here; SwD found it inferior). The σ-gated banded-alignment parameters were taken from one shot at pre-registered values, not searched.

7 Conclusion

When does training on downscaled images yield the same gradients? Not when the noise has masked the frequencies the coarse grid loses. That spectral answer was advanced by prior work for inference-time substitution and had not previously been confronted with the training gradient; transported to the training question, even in its strongest tolerance-parameterized form it is a model of only one part of one branch of what demotion perturbs, and the measurements reject it at every point of structural divergence: the mildest route's crossover sits at 3.5× the equal-power prediction, no tolerance reproduces the map, and a 2×-downscaled latent is certifiably unsafe at every noise level, because the network function itself is grid-calibrated, partly through its positional coordinate system and partly through an irreducible dependence of the computation on token count. The measured answer has two conditions, matching the two terms of the account: the target grid must be large enough in *absolute token count* that its graph floor sits below certification resolution, and the noise level must exceed a per-route boundary where the ratio-governed data term dies. On our model that means 1024→896 at $\sigma > 0.5$ and 1280→1024 at $\sigma \gtrsim 0.75$, and a 2× downscale at no noise level. The answer is predictive, not merely descriptive: the exact angular form of the account, with its two measured governors, predicts held-out routes at ~0.07–0.09 RMSE, and its floor law correctly predicted both held-out floors. Getting there required treating the question as a metrology problem:

naive finite-draw estimation manufactures exactly the token-count-ordered floor signature under test, half the intermediate route's raw floor was estimator bias, and our own headline claims were re-derived, and in one case softened, under the correction. On the one route our corpus makes safe, the trainer realizes the map's projected ceiling (-14.6% wall-clock at fixed steps) with a sample-level footprint within the seed lottery; quality non-inferiority at production scale remains the open gate. We offer the account and its instrument (per-σ-bin gradient agreement with a redraw floor, a re-encoding control, per-arm self-floors, and a stated certification resolution, criteria frozen before data) as the cheap, general test that any future "train part of the time at lower resolution" proposal should pass before it is believed; without the debiasing, that test manufactures the very effect it looks for.

Reproducibility statement

The probe scripts (`run_sigma_probe.py`, including the `--self_floor` and `--draw_sweep` debiasing modes; `run_prior_distance.py`, `rapsd.py`, `compare_ckpt_dw.py`), the held-out validation and refit analyses (`e5_holdout.py`, `e5_refit.py`), the trainer wiring (`--sigma_lowres`), and the frozen pre-registration documents (including the debiasing campaign's decision rules) are in the public repository at `https://github.com/sorryhyun/anima_lora` (project line `project/sigma_lowres/`). The debiased verdict runs' complete per-image records (`result.json` + `per_image.jsonl`) ship in-repo under `project/sigma_lowres/paper_bench/runs/`; the earlier raw-instrument run archives are excluded by the repository's results ignore rule and will be published as a companion tarball **[pending]**; until then, raw-run numbers are reproducible from the scripts but their original records are not themselves public. Paired training runs use a `--deterministic` mode (deterministic attention backward included) verified bit-identical across twin runs; finite-draw cosines are kernel-path-sensitive, so all cosine pairings are computed within a single run (Appendix C).

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A Derivation of the branch decomposition (Eq. 3)

Fix a query (image, caption), a σ, and an arm a. Per draw ϵ, the adapter gradient of the $\frac{1}{2}$-scaled loss is the product $g_a(\epsilon) = J_a(\epsilon)^\top r_a(\epsilon)$ of §3.2. Split each factor into its noise mean and fluctuation, $J_a = \bar{J}_a + \tilde{J}_a$ and $r_a = \bar{r}_a + \tilde{r}_a$ with $\mathbb{E}_\epsilon[\tilde{J}_a] = 0$ and $\mathbb{E}_\epsilon[\tilde{r}_a] = 0$; the population mean gradient of arm a is then

$$\bar{g}_a \;=\; \mathbb{E}_\epsilon\big[J_a^\top r_a\big] \;=\; \bar{J}_a^\top \bar{r}_a \;+\; \mathbb{E}_\epsilon\big[\tilde{J}_a^\top \tilde{r}_a\big]. \tag{9}$$

Write the demoted arm's mean factors as native plus a perturbation, $\bar{J}_{\text{dem}} = \bar{J} + \Delta J$ and $\bar{r}_{\text{dem}} = \bar{r} + \Delta\bar{r}$ (bars without an arm subscript denote the native arm). Subtracting Eq. 9 across arms and expanding the product,

$$\delta g \;=\; \bar{g}_{\text{dem}} - \bar{g}_{\text{src}} \;=\; \underbrace{\bar{J}^\top \Delta\bar{r}}_{\text{data branch } B_e} \;+\; \underbrace{\Delta J^\top\, \bar{r}}_{\text{graph branch } C_e} \;+\; \underbrace{\Delta J^\top \Delta\bar{r} \;+\; \mathbb{E}_\epsilon\big[\tilde{J}_{\text{dem}}^\top \tilde{r}_{\text{dem}}\big] - \mathbb{E}_\epsilon\big[\tilde{J}_{\text{src}}^\top \tilde{r}_{\text{src}}\big]}_{\text{remainder } R_e}. \tag{10}$$

This is an exact identity: no linearization is involved. "First-order" in the main text refers only to the content of R_e: it collects every term that is a product of two perturbations ($\Delta J^\top \Delta\bar{r}$) or a difference of noise-covariance terms (the last two). Eq. 3 is Eq. 10 with the mean-Jacobian bar suppressed in the notation.

One remark delimits the identity's literal scope, expanding the qualification stated in §3.2. When the two arms share a grid (as for the re-encoding control), every operation above is literal. Across grids, J_{src} and J_{dem} act on different token counts, so the differences ΔJ and $\Delta\bar{r}$ require a fixed identification of the two output spaces (e.g. spectral zero-padding of the coarse grid onto the fine grid's bands); any such identification changes the split between B_e, C_e, and R_e but not their sum δg, which lives in the shared adapter-parameter space regardless. This is why the main text defines the branches *operationally* (as the two independent perturbation directions in adapter-parameter space, isolated by the designated probes of §4.3) and treats Eq. 3 as fixing what each branch collects rather than as a computable matrix formula.

B Derivation of the angular forms (Eqs. 4 and 5)

Both angular forms follow from Eq. 1 and the split of δg by the native direction; no property of demotion enters. Write $g = \bar{g}_{\text{src}}$, $G = \|g\|$, $\hat{g} = g/G$, and $u^\perp = u - (\hat{g}^\top u)\,\hat{g}$ for the component of a vector orthogonal to g (the main text's notation); decompose $\delta g = G\,\kappa_\parallel\,\hat{g} + \delta g^\perp$ with $\kappa_\parallel = \hat{g}^\top \delta g / G$ and $\kappa_\perp = \|\delta g^\perp\|/G$.

The exact form. The two components are orthogonal, so $\|g + \delta g\|^2 = G^2(1+\kappa_\parallel)^2 + G^2\kappa_\perp^2$ and

$$\cos\big(g,\; g+\delta g\big) \;=\; \frac{\hat{g}^\top(g+\delta g)}{\|g+\delta g\|} \;=\; \frac{1+\kappa_\parallel}{\sqrt{(1+\kappa_\parallel)^2+\kappa_\perp^2}} \;=\; \frac{1}{\sqrt{1+\kappa_{\text{eff}}^2}}, \qquad \kappa_{\text{eff}} = \frac{\kappa_\perp}{1+\kappa_\parallel}, \tag{11}$$

valid whenever $1+\kappa_\parallel > 0$ (the perturbation does not reverse the gradient direction); $d_e = 1 - \cos$ is Eq. 4. Since $1 - \cos(\hat{g}_1, \hat{g}_2) = \frac{1}{2}\|\hat{g}_1 - \hat{g}_2\|^2$ for unit vectors, starting from either expression in Eq. 1 lands on the same result.

The quadratic limit. For small perturbations, $1 - (1+\kappa_{\text{eff}}^2)^{-1/2} \;=\; \frac{1}{2}\kappa_{\text{eff}}^2 + O(\kappa_{\text{eff}}^4)$ and $\kappa_{\text{eff}}^2 \;=\; \kappa_\perp^2\big(1 - 2\kappa_\parallel + O(\kappa_\parallel^2)\big)$, so

$$d_e \;=\; \frac{\|\delta g^\perp\|^2}{2G^2} \;+\; \underbrace{O(\kappa_\perp^2\kappa_\parallel) + O(\kappa_\perp^4)}_{\text{beyond-quadratic}}. \tag{12}$$

The four-term expansion. Substituting $\delta g \;=\; B_e + C_e + R_e$ (Eq. 10) into the leading term and expanding the square,

$$\|\delta g^\perp\|^2 = \|B_e^\perp\|^2 + \|C_e^\perp\|^2 + 2\,\langle B_e^\perp,\, C_e^\perp\rangle + \Big(2\,\langle (B_e{+}C_e)^\perp,\, R_e^\perp\rangle + \|R_e^\perp\|^2\Big),$$

whose first three terms, divided by $2G^2$, are exactly the data share S_e, the graph share F_e, and the projected interaction I_e of Eq. 5. The remainder R'_e of Eq. 5 therefore has an exact content: the R_e-bearing terms above, the beyond-quadratic terms of Eq. 12, and the parallel-component correction, each suppressed by an extra power of perturbation size relative to the retained terms.

What is not derived. The loading $\|\delta g^{\perp}\| = c_e\, m(\sigma)$ (that the orthogonal gradient mismatch is proportional to the residual mismatch with a σ-independent route gain) is an empirical ingredient (the data-branch factorization at the amplitude level), fit and tested held-out in §4.6. The geometry above fixes only how a given δg is read into gap units.

C Instrument details

Estimator. Per image, arm, and σ-bin, adapter gradients are accumulated over D stratified draws and flattened; cosines are taken between accumulated vectors. Per-bin cosines at D=8 are not comparable *across* bins (the floor drops where $\|g\|$ is small); the gap subtraction against the same-bin floor is the valid read. Split-half reliability over images of each bin-mean curve is mandatory (observed 0.73–0.88 on verdict runs); a per-image variant of the same quantity has split-half reliability ≈ 0 and is not used.

Controls. The re-encoding arm prices the VAE decode–encode round trip that demotion necessarily pays; in pre-debiasing runs its gap served as the validity band (±0.04; observed $|\cdot| \leq 0.054$ across all bins of the main runs, per-module-group ≤ 0.045 in the split runs), a role superseded by the paired debiased read of §4.1. The redraw floor prices finite-draw noise. Density-weighting the uniform bins by the trainer's σ-density reproduces earlier σ-marginalized measurements from the same instrument family (consistency check).

Debiasing (§3.5), implementation. Under `--self_floor`, every arm (re-encoding and each demote variant) runs a second independent draw set from a disjoint seed stream, giving per-arm self-cosines alongside the native floor; Eq. 8 is then computed per image and bin, and the map read is the paired excess $\mathrm{gap}_{e,i} = \widehat{\mathrm{gap}}_{e,i} - \widehat{\mathrm{gap}}_{\mathrm{reenc},i}$ with non-finite values (negative self-cosines under the square root at low D) and $|\mathrm{gap}_{e,i}| > 1.5$ trimmed (0–5 of 40 images per cell). Under `--draw_sweep`, draw prefixes are nested ($D = 64$ contains the $D \leq 32$ sets; prefix sums, no extra forwards) and $\mathrm{gap}(D) = \mathrm{gap}_{\infty} + c/D$ is fit on route means with a bootstrap CI over images. Finite-draw cosines are kernel-path chaotic: twin runs sharing a warm compiler kernel cache agree to $|\Delta \cos| \leq 0.015$, while a run compiled to a different kernel set lands up to ≈ 0.3 away at D=2, so gap/floor/self-floor pairings are not compared across processes, a guarantee the single-run instrument provides by construction; a `--deterministic` mode (bit-exact across twin runs) covers the paired training A/Bs.

Endpoint and x-zero modes. The σ=1 endpoint bin feeds input exactly ϵ (the input-mediated data term vanishes by construction). The x-zero mode zeroes x in input *and* target on every grid, keeping captions and exact demoted latent shapes; with the $(1-\sigma)x$ term absent, low-σ bins are off-manifold and the σ=1 read is primary. In that regime the residual is $\approx -\hat{x}_{\mathrm{prior}}$, so "graph-dominated" includes the model's grid-conditioned prior; the prior-distance probe (§4.3) subsequently dissociated the prior from the floor's route ordering.

Held-out validation and refit (§4.6), sources. The fit consumes the debiased paired curves of the 1024-tier verdict run, the 1280-tier curves (pre-debiasing paired reads; caveat recorded in §6), the route-uniform amplitude $m(\sigma)$ from the residual probe, and each run's own bin-mean gradient norm $\|\bar{g}_{\mathrm{src}}(\sigma)\|$ (gap and norm always from the same process, per the kernel-path rule). The shared-power fit scans $p \in [1, 2]$ jointly across fit routes with per-route weighted least squares; the exact-link fit is a per-route grid with profile-likelihood uncertainties on c_e. Oracle baselines are quadratic-in-σ fits on the held-out data itself (the noise ceiling for a 3-dof curve); the spectral baseline is gap = 0 on the spectral account's committed region $\sigma > \sigma_{\mathrm{eq}}$.

Bootstrap bands and the leave-896-out lane (Fig. 2). The bands are a full-pipeline image bootstrap (B=1000, fixed seed; 981 draws kept): resample images with replacement within each run (N=40 / N=24), re-bin the paired curves, refit the exact-link (c_e, Floor_e) on every fit route, re-derive

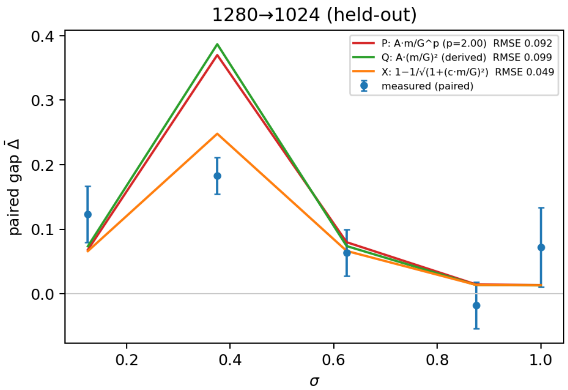


Figure 4: The second held-out route, 1280⟶1024, under all three functional forms for the data term of §4.6 (fitted power, derived quadratic, exact angular link), predicted from governors fitted on other routes. This is the one route where the forms visibly separate: the derived quadratic systematically overshoots the mid-σ peak; the exact angular link's saturation removes the overshoot (RMSE 0.049 vs ≈ 0.095), which is why the exact link headlines in Fig. 2.

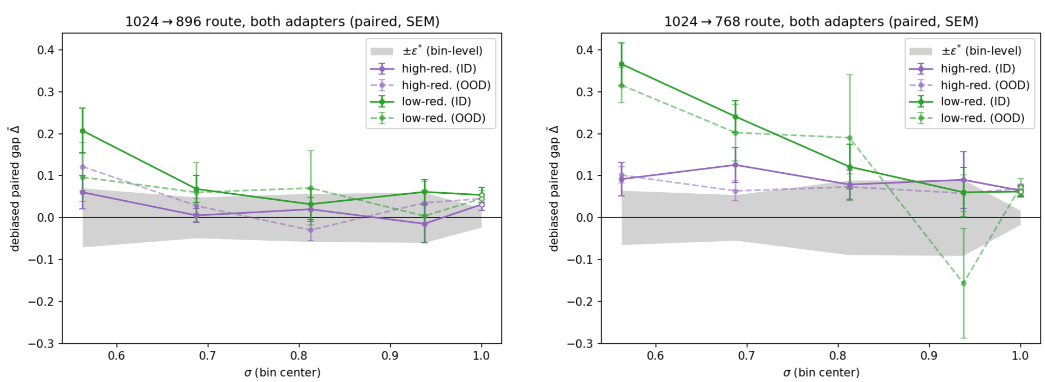
the ratio governor and floor law, re-predict. $\bar{m}$ and the bin-mean $\|\bar{g}_{\mathrm{src}}(\sigma)\|$ are run-level instruments with no per-image decomposition and are held fixed. The replica includes the committed floor law's positive-floor filter (Floor > 0.005): in 41% of draws the resampled 1120 floor clears the filter and the law re-anchors on the $(896, 1120)$ legs rather than $(512, 896)$, branch variability the bands therefore contain. The 896 curve in Fig. 2 is the post-hoc leave-896-out lane under the exact link (c from the ratio-twin 1120 alone $= 0.142$ vs 0.150 fitted on 896 itself; floor law through 512+1120, whose fragile Floor_{1120} leg enters the log fit clamped positive (36% of bootstrap draws clamp), giving $F(3012) = +0.012$ vs measured $+0.019$ $[+0.010, +0.030]$; curve RMSE 0.073), the X-form counterpart of the quadratic-form addendum (§4.6).

Interventional B/C ledger (§4.6), implementation. Routes $1024 \to \{896, 768, 512\}$, $\sigma \in [0.5, 1.0]$ in four bins plus the σ=1 endpoint bin, D=8 draws per bin, N=24 images, deterministic mode, two independent draw sets per arm. Each arm's draw-summed adapter gradient is stored per bin; the interventional split is $B = \bar{g}_{\mathrm{repromote}} - \bar{g}_{\mathrm{src}}$ (the data intervention at fixed native graph: the demote–re-promote arm's input passed through the downscale⟶upscale⟶encode pipeline but is evaluated on the native grid) and $C = \bar{g}_{\mathrm{demote}} - \bar{g}_{\mathrm{repromote}}$ (the graph intervention at fixed demoted data). The quadratic shares S, F, I, and $\rho = \cos(B^{\perp}, C^{\perp})$ use cross-draw-set inner products to cancel shared draw noise; the debiased estimates are not confined to $[-1, 1]$ (hence $\rho = -1.24$ at one low-amplitude bin). The exact counterfactual angles $h(B)$, $h(C)$, $h(B{+}C)$, with $h(X) = 1 - \cos(\bar{g}_{\mathrm{src}}, \bar{g}_{\mathrm{src}} + X)$, are computed on the raw arm means. The re-encoding proxy check re-references the data leg against the control arm, $B_{\mathrm{reenc}} = \bar{g}_{\mathrm{repromote}} - \bar{g}_{\mathrm{reenc}}$, and compares $|B^{\perp}_{\mathrm{reenc}}|$ to $|B^{\perp}|$ per bin.

Spectral-account computation. Table 2 evaluates Eq. 7 at the demoted grids' Nyquist frequencies $f_{\mathrm{cut}} = \frac{1}{2}\, e/e_0$ on the N=40 mean radially-averaged latent power spectrum (64 radial bins; P at the 896/768/512 cuts = 0.025/0.029/0.065). The equal-power slice $\delta = (1 + P_{\omega})/2$ recovers $\sigma_{\mathrm{eq}} = \sqrt{P}/(1 + \sqrt{P})$ exactly. Across $\delta \in [0.001, 0.5]$ the mildest–harshest spread remains below 0.13, and the 1024⟶896 versus 896⟶768 boundaries (cut frequencies 0.4375 vs 0.4286) separate by no more than 0.01. The parameter-free anchor δ_{reenc} of §4.2 is measured by the same estimator: RAPSD of the re-encode arm's latent error (fresh resize–encode minus cached latent, the gradient probe's own control chain) on the same probe set, interpolated at each route's cut frequency. Measured (N=40, 64 radial bins): error variance 2.7×10^{-5} against latent variance 0.419; $D(f_{\mathrm{cut}}) \approx 1.0 \times 10^{-5}$ at all three cuts (p10–p90 over images within $[1, 2] \times 10^{-5}$), giving $t^{*}_{\mathrm{reenc}} = 0.98/0.98/0.99$ for 896/768/512.

Table 5: The interventional B/C ledger behind §4.6: per (route, σ-bin) orthogonal amplitudes (units of $\|\bar{g}_{\text{src}}\|$), interaction geometry, the cross-set-debiased quadratic shares, and the exact counterfactual angles. In-window S+F+I is sign- and localization-accurate but underpredicts $h(B{+}C)$ (small-perturbation truncation out of domain); magnitude claims in the text use $h(\cdot)$.

route	σ	$\lvert B^{\perp}\rvert$	$\lvert C^{\perp}\rvert$	ρ	S	F	I	$h(B)$	$h(C)$	$h(B{+}C)$
896	0.563	0.55	0.59	−0.95	0.103	0.115	−0.206	0.107	0.206	0.037
	0.688	0.42	0.39	−0.91	0.036	0.059	−0.085	0.083	0.077	0.062
	0.813	0.20	0.23	−0.94	0.016	0.019	−0.033	0.022	0.027	0.010
	0.938	0.10	0.10	−1.24	0.002	0.003	−0.006	0.007	0.006	0.003
	1.000	0.05	0.06	−0.71	0.001	0.001	−0.001	0.002	0.003	0.002
768	0.563	0.57	0.68	−0.93	0.139	0.200	−0.309	0.107	0.346	0.088
	0.688	0.56	0.57	−0.93	0.100	0.136	−0.218	0.106	0.242	0.098
	0.813	0.35	0.42	−0.93	0.049	0.067	−0.105	0.059	0.110	0.035
	0.938	0.14	0.17	−1.02	0.005	0.009	−0.014	0.010	0.033	0.008
	1.000	0.07	0.09	−0.62	0.001	0.004	−0.003	0.003	0.014	0.013
512	0.563	0.60	1.09	−0.72	0.166	0.403	−0.372	0.085	0.854	0.466
	0.688	0.63	0.88	−0.86	0.153	0.332	−0.388	0.107	0.623	0.305
	0.813	0.57	0.72	−0.87	0.152	0.226	−0.323	0.131	0.285	0.126
	0.938	0.25	0.29	−0.96	0.026	0.036	−0.058	0.028	0.201	0.037
	1.000	0.13	0.15	−0.88	0.007	0.010	−0.015	0.007	0.379	0.098



(a) 1024→896 (the certified route).

(b) 1024→768.

Figure 5: The Fig. 1a recipe repeated on two controlled adapters trained on opposite style clusters, one panel per route (N=48 stems per adapter, D=8/bin, high-σ window; same paired debiased estimator and trim; open markers the σ=1 endpoint bin; axes shared across panels). Unlike the main-text figures, color encodes the *adapter* (high- vs low-redundancy cluster); solid curves are in-distribution stems, the adapter's own style cluster (n=36: trained-on, held-out, and unseen-artist cells), and dashed curves are out-of-distribution stems from the opposite cluster (n=12). Gray band: bin-level $\pm\varepsilon^{*}$ per Eq. 2, median of the two runs' full-N SEMs for the panel's route. The map replicates: on the certified route all four adapter×distribution curves reach the band inside the window, on 768 all four sit above it until the top, and within each panel the dashed OOD curves track their solid ID counterparts: the adapter×probe-style interaction is not visible at this resolution.

D Controlled-adapter replication of the verdict map

The one-adapter caveat of §6 pre-registered a controlled 2×2 factorial: two plain-LoRA checkpoints trained under the shipped probe adapter's frozen recipe on opposite style clusters (the top and bottom 12 artists by median latent redundancy, hereafter the high-redundancy, visually flat, and low-redundancy, heavily textured, clusters), with stem-level membership manifests frozen before training (124 training images each, identical seeds). Each adapter was probed on 48 stems spanning four membership cells (N=12 each): trained-on, held-out images of trained artists, unseen artists of the same cluster, and unseen artists of the opposite cluster, with the cross-cluster stems shared between the two runs so cross-adapter contrasts read paired per stem. Figure 5 repeats the verdict-map recipe per route, overlaying both adapters.

Three reads, stated at the bin-mean level (per-cell tables reproduce from the archived per-image rows). First, the (route, σ) shape replicates on both checkpoints (Fig. 5). Second, the factorial contrasts are null at instrument resolution: the probe-style main effect and the adapter×probe-style interaction (the pre-registered verdict quantity) are both bounded below the resolvable bin-mean effect (raw-gap paired interaction -0.022 ± 0.027 at 896, -0.015 ± 0.059 at 768), and the membership contrast (trained-on versus held-out) does not replicate in sign across the two adapters. The map is therefore adapter- and content-agnostic on this model at our resolution, supporting the calibrate-once-per-model reading of §5.1. Third, the one quantity that is *not* checkpoint-invariant is the absolute redraw-floor level: in-window floor cosines sit at ≈ 0.73 for the high-redundancy-cluster adapter versus ≈ 0.50 for the low-redundancy-cluster adapter, uniformly across all four membership cells including the opposite-cluster stems. The floor's level is a property of the checkpoint (its gradient stochasticity), while the safety map built on top of it is not.

E Historical (pre-debiasing) tables

The tables in this appendix are the raw-estimator record: retained for continuity and for the reads defended in §4.1 (one-signed bias; paired same-grid contrasts), not cited as debiased evidence.

Table 6: **Raw estimator** (historical record; superseded by Table 1): demotion gap by σ-bin, 1024-tier natives (N=40, bin-mean SEM ~0.02, re-encoding control $|\mathrm{gap}_{\mathrm{reenc}}| \leq 0.054$ everywhere, split-half reliability 0.73–0.83). Bold = within the re-encoding band. Raw values understate mid-σ gaps (the floor attenuates the read) and overstate endpoint floors (single-estimate variance bias); the debiased map corrects both. The $\cos_{\mathrm{floor}}$ and $\|g\|$ rows are the estimator context plotted in Fig. 1b.

σ bin center	0.06	0.19	0.31	0.44	0.56	0.69	0.81	0.94
gap_{896} (ratio 0.875)	.110	.162	.148	.137	**.048**	**.030**	**.030**	.053
gap_{768} (ratio 0.75)	.144	.216	.208	.223	.164	.163	.115	.063
gap_{512} (ratio 0.5)	.348	.410	.355	.469	.430	.391	.289	.296
$\cos_{\mathrm{floor}}$	.84	.65	.51	.65	.70	.77	.80	.83
$\|g\|$ (native)	2.6	0.5	0.4	0.5	0.7	1.1	2.0	9.3

Table 7: Raw endpoint reads (D=16, N=40) beside their debiased draw-limit values (Table 3), the debiasing's revision record: the 768 floor halves, the 896 floor surfaces.

route	raw endpoint (D=16)	endpoint $\widehat{\mathrm{gap}}_\infty$ [95% CI]
native self-floor (cosine)	0.85	1.005 [0.994, 1.016]
re-encoding control	−0.038	−0.003 [−0.017, +0.008]
1024→896	−0.009	+0.019 [+0.010, +0.030]
1024→768	+0.127	+0.056 [+0.043, +0.071]
1024→512	+0.326	+0.304 [+0.197, +0.424]

Table 8: Iso-severity probe (**raw estimator**): gap by σ-bin on the 1280-tier cache (N=24, 4 draws/bin) against the corpus 1024→896 curve. The ratio-matched routes coincide within ~1.4 combined SEM at every bin despite a 1.6× difference in absolute target size; the same-native harsher-ratio route separates. *896's canonical 16-draw endpoint is −0.009 raw, +0.019 debiased (Table 3).

σ	0.125	0.375	0.625	0.875	1.0
re-encoding control	+.047	−.005	+.048	+.020	−.012
1280→1120 (ratio .875)	.129	.110	.054	−.012	−.007
1280→1024 (ratio .80)	.170	.178	.111	.002	.061
1024→896 (ratio .875)	.132	.076	.077	.049	.100*

Table 9: Ratio-transfer run (**raw estimator**): 896-tier natives (N=40), pre-registered bar "gap$_{768}$ within the re-encoding band at $\sigma \geq 0.5$": **fails** (residual 0.06–0.12, ~2× the 1024→896 plateau; ~half of it is expected estimator bias, §4.1; the separation is independently confirmed by the debiased same-grid floors, §4.3).

σ bin	0.06	0.19	0.31	0.44	0.56	0.69	0.81	0.94
896→768	.114	.209	.168	.143	.124	.056	.092	.061
896→512	.318	.372	.308	.340	.320	.280	.329	.217

Table 10: Route-uniformity (in amplitude) of the mean prediction residual: cross-grid excess of $\bar{r} = \mathbb{E}_\epsilon[\hat{v} - (\epsilon - x)]$ over split-half floors (relative L_2; same-grid re-encoding control ≤ 0.02 everywhere). The three main routes coincide within $\sim \pm 0.02$ at every σ while their gradient floors span 0 to 0.3. This is a norm read; the direction-resolved follow-up (§4.4) shows the underlying mismatch directions are image- and route-specific.

σ	0.125	0.375	0.625	0.875	1.0
1280→1024	.885	.825	.715	.465	.360
1024→896	.862	.808	.706	.459	.378
896→768	.860	.814	.715	.466	.393
768→512	.897	.872	.767	.508	.435

Table 11: Depth localization of the floor (**raw estimator**): mean per-block gap at σ=1, early = blocks 0–9, late = 14–27, for the endpoint (ep) and x-zero (xz) probes. Early blocks carry ~3× the late-block gap. The apparent content share (ep − xz) is a late-block minority effect in raw units; the debiased draw-limit comparison finds ep = xz within CI at the whole-adapter level (Table 3).

route	ep early	ep late	xz early	xz late
1024→512	.357	.223	.351	.125
1024→768	.164	.085	.121	.033
1024→896	.023	.015	.044	.012

Table 12: Positional-interpolation intervention at the σ=1 endpoint (N=40, 16 draws; **raw** arm values; the paired Δ column is a same-grid contrast in which the finite-draw bias cancels to first order): exact phase-geometry alignment erases the mild-route floor and ~30% of the harsh-route floor.

route	plain (SEM)	PI-aligned (SEM)	paired Δ (SEM)	improved
1024→896 (control)	−0.021 (.041)	−0.040 (.040)	—	—
1024→768	+0.080 (.048)	−0.001 (.039)	+0.081 (.031)	78%
1024→512	+0.320 (.058)	+0.224 (.056)	+0.096 (.039)	70%